ANALYSIS · FOOTBALL ECONOMICS

What money can't buy

A football performance measurement framework

2013/14 – 2024/25

Ordered logit model · Monte Carlo simulation · 50,000 iterations per season

EXECUTIVE SUMMARY

What money can't buy

A football performance measurement framework

Football is an industry that generates more managerial decisions than almost any other — and fewer of them are grounded in systematic evidence. Coaches are hired on reputation, fired on a five-game run, and evaluated against a league table that conflates the quality of management with the size of the wage bill. This analysis attempts to do something more rigorous: using every match played across twelve complete seasons in La Liga and the Premier League — 9,008 matches, 480 team-seasons, two of the most watched leagues in the world — it builds a formal model of what results should look like given each club's resources, and measures everything above and below that line. The findings are often counterintuitive. The implications are direct.

Three recommendations for football clubs

1. Evaluate performance against the budget, not the table

The league table measures outcomes, not management. A club finishing fifth with the third-largest wage bill is underperforming; a club finishing eighth with the fifteenth-largest is overperforming. The two situations are indistinguishable in the table and require opposite responses. Any serious performance evaluation must begin by setting a wage-based baseline — what the money predicts — and measuring every result above or below that line. The Managerial Success Score (MS) developed in this analysis does precisely that: it combines the direction, magnitude and statistical confidence of a club's multi-season residual into a single number, allowing managers to be ranked not against each other but against the potential each was given.

2. Invest in managerial continuity — it is the single most consistent differentiator

Across 127 five-year windows in both leagues, the clubs that simultaneously sustained high overperformance and achieved strong payroll growth share one attribute without exception: a coherent, multi-year coaching project. Guardiola at Man City, Klopp at Liverpool, Simeone at Atlético, Pochettino at Tottenham, Alguacil at Sociedad, Emery at Villarreal. This holds across all budget levels — from Getafe at €27M to Man City at £156M. Budget size is not the differentiator. The clubs that failed — Valencia, Celta, Crystal Palace, Man United — all had managerial revolving doors of four or more coaches in eight seasons, regardless of their wage bill. A Fisher exact test on the association between managerial continuity and quadrant membership yields $p<0.03$ in both directions.

3. Take a multi-year strategic view — one season is rarely enough signal

A single season of overperformance has a near-zero correlation with the next season's payroll growth; five consecutive years of overperformance explains 47% of payroll growth two years later in La Liga and 23% in the Premier League. The implication is direct: boards that fire a manager after one difficult season — particularly one following a period of investment and structural change — are acting on noise, not signal. The MS score is specifically designed to separate signal from noise: a score above +70 or below −70 reflects a multi-season residual large enough to rule out luck at conventional significance levels. Reaching that threshold takes time, and dismantling it prematurely destroys the feedback loop between sustained overperformance, commercial growth and competitive reinvestment.

The same logic applies within a season. Simulating 50,000 seasons per club-season reveals that a top-quartile club going winless across five games is a statistically normal event in roughly 30% of seasons. Three consecutive defeats is the median worst run for every budget tier. A rolling bad

run only becomes a genuine signal — below the 5th percentile of simulations — at a threshold most boards never wait for. The two-step diagnostic developed in Chapter 6 (run percentile + season-level MS score) allows any board to determine, in real time, whether a difficult patch is noise or structural deterioration. The Xabi Alonso case at Real Madrid in January 2026 illustrates the point precisely: fired while La Liga overperformance stood at +8.3 points and the worst 5-game run sat at the 89th percentile of expectations.

The supporting evidence

The model. We analyse every match — all 9,008 from twelve complete seasons in La Liga and the Premier League — as a function of the wage ratio between the two clubs. The pattern is clear and monotonic: the greater the financial advantage, the higher the win rate. An ordered logistic regression converts this relationship into a formal model that explains 68% of the variance between teams at season's end in La Liga and 52% in the Premier League, with mean absolute errors of ±10 and ±13 points respectively. With a single variable. What the model does not capture — the manager, injuries, dressing-room cohesion — is precisely what we want to measure.

Overperformance is structural. Once the budget is controlled for, each team's accumulated overperformance becomes a signature. Villarreal and Atlético de Madrid have each delivered over +60 points above expectation across twelve La Liga seasons; Liverpool and Man City +156 and +138 in the Premier League. At the negative extreme, Valencia have accumulated −80 points; Man United −109 while holding the league's largest wage bill for much of the period. The probability that the extreme cases are the product of chance is below 10%. The Managerial Success Score (MS) — combining direction, magnitude and statistical confidence — makes this rigorous: scores above +70 are structural signals of management quality; scores below −70 are structural signals of destruction of value.

Overperformance drives budget growth. Using a five-year lagged regression (W=5, k=2), sustained overperformance explains 47% of subsequent payroll growth in La Liga and 23% in the Premier League — and does so regardless of which metric is used to measure overperformance (MS score, normalised %, raw points all yield statistically indistinguishable R^2). Boards respond to the scoreboard, not the underlying process. One lucky season is rewarded as much as one earned by tactical superiority. This makes managerial continuity even more valuable: it is the mechanism by which overperformance compounds into structural budget advantage.

Two leagues, one pattern. La Liga is more deterministic — the champion has always come from the top two or three wage bills, and the overperformance-to-budget feedback is stronger (R^2=0.47 vs 0.23 in the Premier League). The Premier League is more chaotic: champions have come from wage rank #1 down to #16, and the dispersion of overperformance is almost double. A plausible hypothesis is that greater financial balance does not produce more predictability but more noise — when wage bills are more similar, management carries more relative weight. The data are consistent with this, though confirming it would require controlling for additional variables.

Money buys probabilities, not certainties. When one team doubles the rival's wage bill, the home side's win probability rises by 12 percentage points in La Liga and 17 in the Premier League. The relationship is real, persistent and measurable. But 53% of payroll growth variance remains unexplained after controlling for performance, and 32–48% of season-end point variance is unexplained after controlling for wages. The space between the model's prediction and what actually happens is not residual noise — it is the space where management creates or destroys value. The purpose of this analysis is to measure that space precisely enough to act on it.

Most bad runs are variance. Simulating 50,000 seasons per club-season — preserving each team's actual fixture list and wage ratios — shows that a top-quartile club going winless across five games is expected by chance alone in roughly one in three seasons. Three consecutive defeats is the median worst run for every budget tier. A rolling bad run is only statistically unusual below the 5th percentile of simulations, a threshold that most managerial dismissals never reach. The two-dimensional diagnostic map introduced in Chapter 6 — combining rolling-run percentile with season-level MS score — allows any board to distinguish in real time between a difficult patch and structural deterioration. The analysis finds that the majority of observable bad runs in both leagues sit in the normal range; the managerial decisions made in response to them are, by definition, responses to noise.

SECTION 1

Introduction

How Do You Measure Success in Football Club Management?

Assessing the performance of a manager or sporting director is one of the most complex questions in modern football. Unlike other sectors, sporting results depend on factors that go far beyond management decisions: the fixture list, injuries, luck at key moments, or above all, **the enormous differences in resources between clubs**. A team finishing tenth on £30m may have managed better than one winning on £300m — but raw results cannot distinguish between the two.

The academic literature has addressed this mainly at season level. Works such as *Szymanski & Kuypers (1999)*, *Hall, Szymanski & Zimbalist (2002)* or *Dobson & Goddard (2011)* documented a robust correlation between wage spending and league position. More recent studies confirmed the wage bill as the most powerful predictor of long-term performance across multiple European leagues *(Gelade & Dobson, 2007; Peeters & Szymanski, 2014)*. However, these studies aggregate 38 matchweeks into a single number.

> *What the literature does not provide is a match-by-match view: how does the wage advantage vary by opponent? Is a team performing above or below expectation in each fixture?*

This Analysis

A methodological caveat: this analysis establishes correlations, not causality. Clubs that win earn more, which allows them to spend more. The endogeneity is well documented and unresolved. What this work demonstrates is that the wage bill is the best observable predictor of season performance; it does not claim that spending more causes winning, but quantifies how much can be predicted from that single variable.

In this work we study the influence of squad wage bills in predicting the outcome of each match. The objective is twofold: on one hand, to build a model that assigns realistic probabilities to each outcome **(home win, draw, away win)** as a function of the wage ratio; and second, to use that model as a baseline for identifying which teams consistently outperform or underperform their wage investment.

The analysis covers La Liga and the Premier League across twelve complete seasons (2013/14 to 2024/25). The temporal frame is set by the wage data source: Capology's records for these leagues begin in 2013/14.

Data

We have used two complementary data sources:

- **Match results:** obtained from football-data.co.uk, covering final result, home and away goals, and date, for all La Liga and Premier League matches between 2013/14 and 2024/25. The dataset covers 9,008 complete-season matches, plus the fixtures played so far in the ongoing 2025/26 season.
- **Wage data:** annual gross wage bill per club and season, obtained from Capology. This source provides the aggregate of all gross wages for the professional squad, including fixed bonuses and supplements, updated annually for each club in both leagues. Capology's historical records for La Liga and the Premier League begin in 2013/14, which defines the analytical window of this study.

For each match, we calculate the **wage ratio** between the away and home teams, expressed as a base-2 logarithm. This transformation has an intuitive interpretation: a value of *+1* means the away team has double the home team's wage bill; a value of *−1* means the home team has double the away team's wage bill; and *0* indicates equivalent wage bills.

Methodology: From Match to Model

From the wage ratio, we estimate an **ordered logistic regression model** — one per league — that relates this ratio to the probability of each outcome. This type of model is particularly suited to ordinal outcomes such as football results, where the natural scale is: home defeat < draw < home win. The model parameters are estimated by maximum likelihood on the full set of matches in each league, yielding calibrated probabilities for any combination of wage bills.

Using the match-by-match probabilities, we build a **Monte Carlo simulation** for each season: we simulate all 380 matchdays in each league 50,000 times, accumulating points for each team based on the random results generated by the model. The result is a complete distribution of expected points per team and season, which serves as a baseline against which to compare the actual points earned.

The difference between actual points and the model's expected points — what we call **overperformance** — is the central metric of this analysis. Sustained positive overperformance indicates that a team is extracting more value from its wage bill than expected; negative overperformance indicates that it is systematically squandering resources.

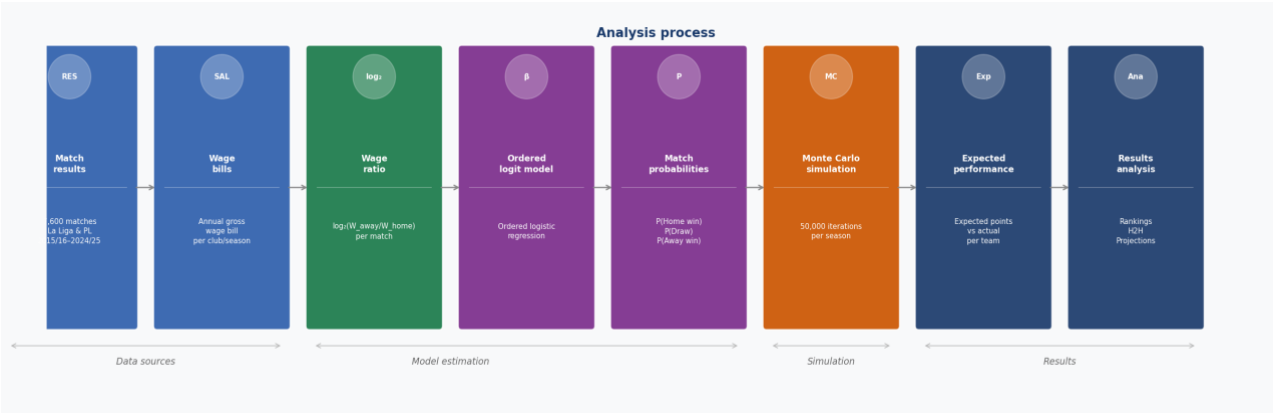

Figure. *Analysis process: from data sources to team performance indicators by season.*

Structure of the Report

The rest of the document is organised as follows:

- **Section 2 — Descriptive Results:** wage bill distribution, inequality between leagues, and the empirical relationship between money and results.
- **Section 3 — The Predictive Model:** specification of the ordered logit model, estimated parameters, predictive capacity at match and season level.
- **Section 4 — Monte Carlo Simulation:** simulation of full seasons, title probabilities and cross-league comparison.
- **Section 5 — Who Outperforms Their Budget?:** cumulative overperformance across twelve seasons, the Managerial Success Score (combining direction, magnitude and statistical confidence), the lagged relationship between sustained overperformance and budget growth, and the club characteristics associated with each quadrant.
- **Section 6 — Fooled by Randomness:** a statistical framework for interpreting bad runs of results; simulation-based reference distributions by budget tier; the diagnostic map of run severity vs season-level overperformance; and a real-world test using Xabi Alonso's dismissal from Real Madrid in January 2026.
- **Conclusions — What the Data Tell Us:** seven conclusions on the relationship between budget, performance, managerial continuity and the statistical interpretation of bad runs, including the strategic implications for club management.
- **Appendix A.1 — Atlético de Madrid:** statistical interpretation of the club's title record and

Champions League qualification probability across twelve seasons.

- **Appendix A.2 — La Liga's Big Three:** head-to-head fixtures and performance against the rest of the league for Real Madrid, Barcelona and Atlético de Madrid.
- **Appendix A.3 — La Liga 2025/26:** projection of the current season with 12 matchweeks to go, the title race, and the teams deviating most from expectation.
- **Annex — Wage Bill and Points Data:** complete tables of squad wage bills and predicted vs actual points for all teams across all twelve seasons.
- **Methodological Note:** description of sources, tools and the use of artificial intelligence in producing this analysis.

References: Szymanski, S. & Kuypers, T. (1999). Winners and Losers. Viking. · Hall, S., Szymanski, S. & Zimbalist, A. (2002). Testing Causality Between Team Performance and Payroll. Journal of Sports Economics. · Dobson, S. & Goddard, J. (2011). The Economics of Football. Cambridge University Press. · Gelade, G. & Dobson, S. (2007). The Development of Club Football in Europe. Journal of the Royal Statistical Society. · Peeters, T. & Szymanski, S. (2014). Vertical restraints in soccer. Economic Policy.

SECTION 2

Descriptive Results

Wage Inequality Is Not the Same in Both Leagues

The first finding emerges before any modelling: **La Liga and the Premier League have radically different wage inequality**. Figure 1 shows the distribution of the wage ratio across both leagues over twelve seasons.

In the Premier League, the distribution is markedly concentrated — with a standard deviation of **1,09** — and there are very few matches in which one team's wage bill is four or more times the rival's: only **5,8%** of fixtures. Most matches involve teams whose wage bills differ by no more than double. The English league is structurally competitive in resources.

La Liga, by contrast, shows a much wider distribution — with a standard deviation of **1,83**, some 68% higher than the Premier League's. In more than one in four matches (**26,5%**), one of the two teams has four times the rival's wage bill. Fixtures such as Real Madrid or Barcelona visiting a newly promoted side are not statistical anomalies: they are a structural part of the fixture list.

> *This difference has direct consequences for predictability. In La Liga, many fixtures are statistically predetermined. In the Premier League, greater wage parity generates more uncertainty — and more surprises.*

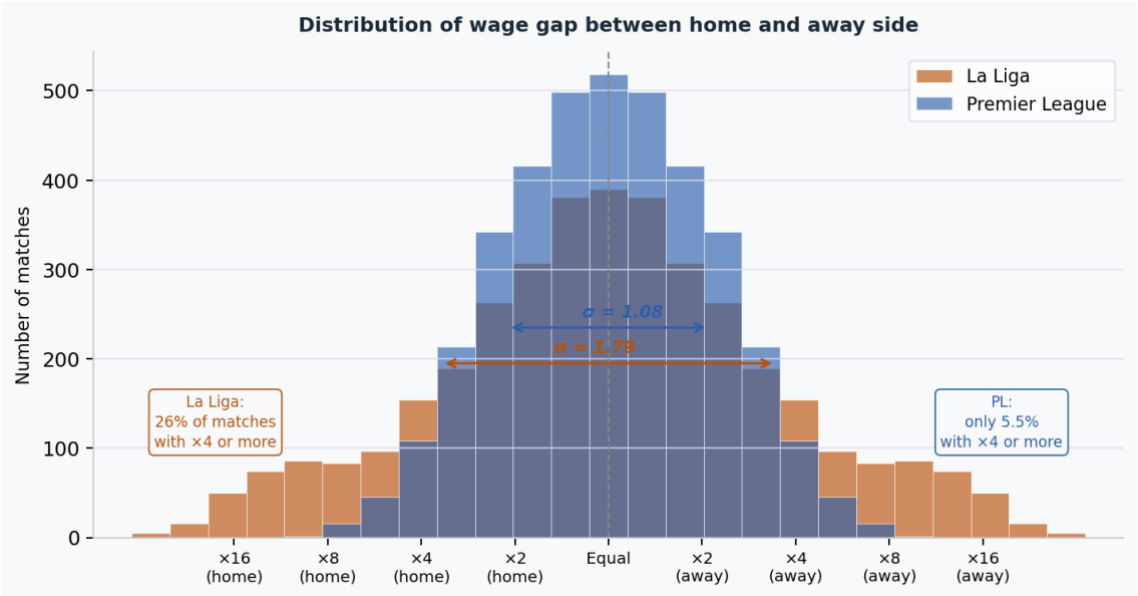

Figure 1. *Distribution of the wage gap between home and away teams in each match, expressed as a multiplier (horizontal axis). The Premier League (blue) shows a markedly more concentrated distribution than La Liga (orange), with a standard deviation of 1.09 vs 1.83.*

Statistical summary of the wage ratio by league

	La Liga	Premier League
Std. dev. of ratio (σ)	**1,83**	1,09
Matches with ×4 or more difference	**26,5%**	5,8%
Matches with ×2 or more difference	**53,9%**	39,0%
Matches between similar teams (±40%)	24,5%	32,9%
P(home win) at equal wage bills	44,8%	44,0%

The Wage Bill as a Predictor of Outcomes: Raw Data

Figures 2 and 3 show the actual — not modelled — results of all 9,008 complete-season matches, grouped by the wage gap between the two teams. The horizontal axis is expressed in **multipliers**: '×2 (away)' means the away team has double the home team's wage bill; '×4 (home)' means the home team has four times the away team's. The bars add to 100% of matches in each group.

The pattern is emphatic. **When the home side has eight or more times the away side's budget, it wins 83% of fixtures**. When the away side holds that advantage, the away win rate reaches 67%. The transition is monotonic: the larger the wage gap, the higher the win rate.

The **draw** shows a particularly revealing pattern: **it peaks precisely when wage bills are most similar** — 29% draws in the 'Equal' bracket — and fall as the gap grows. When resources diverge sharply, the result tends towards clarity.

A second relevant observation is the **home advantage**. In the 'Equal' bracket — matches between teams with equivalent wage bills — the home side wins **42% of fixtures**, compared with 29% for the away side. Without any wage advantage, the home factor still generates a consistent difference.

The cross-league comparison (Figure 3) reveals interesting nuances. In La Liga, mismatched fixtures are far more frequent — as already seen in the distribution — and the outcomes at those extremes are equally emphatic. In the Premier League, the middle ranges concentrate most matches, and **draws are notably less frequent** than in La Liga in almost all brackets: in the 'Equal' bracket, 28% in the Premier League vs 29% in La Liga, with the gap widening in the middle ranges, where La Liga records up to 31% draws. La Liga's higher draw rate is consistent with a more conservative style when teams perceive the match as balanced.

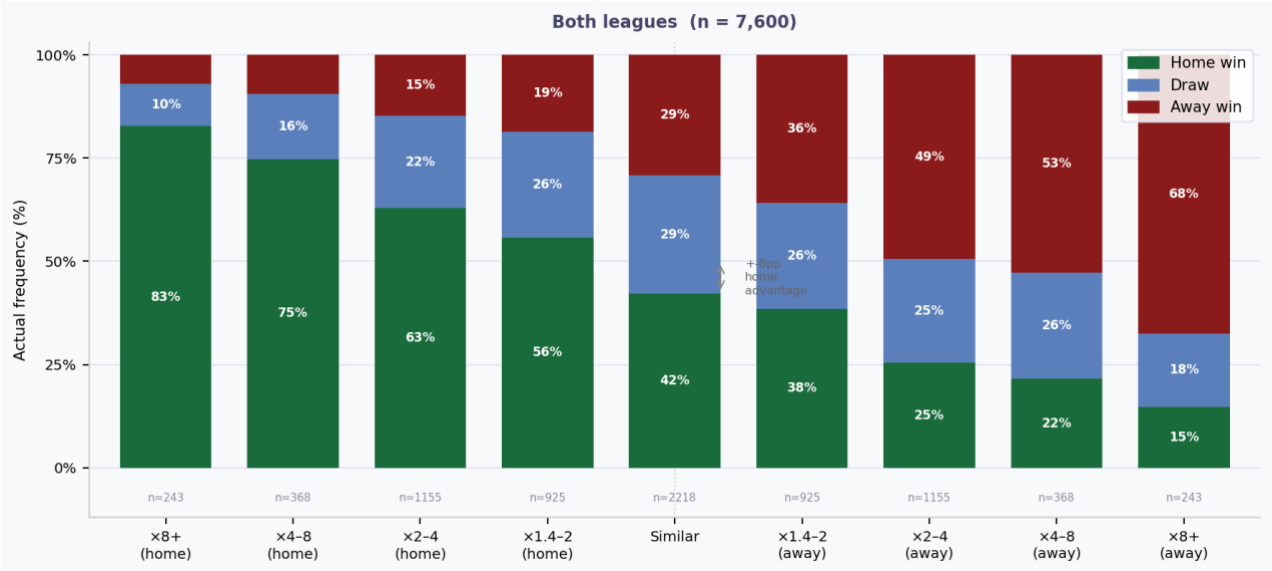

Figure 2. *Actual frequency of home wins, draws and away wins by wage-gap group. Both leagues combined (n = 9,008 matches, full seasons 2013/14–2024/25). Multipliers indicate how many times one team's wage bill exceeds the rival's.*

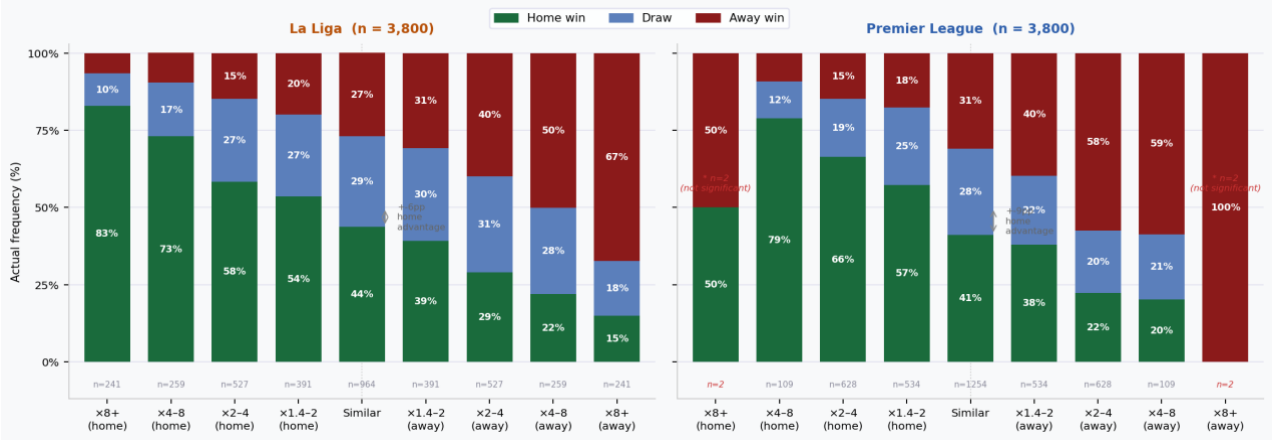

Figure 3. *The same analysis broken down by league. La Liga (left) and Premier League (right). The two extreme cells in the Premier League (×8+) contain only 2 matches and are not statistically representative.*

SECTION 3

The Predictive Model

Model Specification

To convert wage gaps into outcome probabilities, we use an **ordered logistic regression model** — also known as *ordered logit* or *proportional odds model* —. This model is appropriate when the outcome is **ordinal**: match results have a natural ordering (away win < draw < home win) that binary logit or linear regression would not capture.

The explanatory variable is the **log base-2 of the wage ratio** between the away and home teams:

$$x = \log_2 (W_away / W_home)$$

where *W* is the annual gross wage bill for each squad. The log scale has a direct interpretation: $x = +1$ means the away side has double the home side's wage bill; $x = -2$ means the home side has four times the away side's wage bill; $x = 0$ means both wage bills are equivalent.

The model estimates each outcome probability via two nested logistic functions:

$$P(\text{home win}) = 1 - \sigma(\theta_2 + \beta \cdot x)$$
$$P(\text{draw}) = \sigma(\theta_2 + \beta \cdot x) - \sigma(\theta_1 + \beta \cdot x)$$
$$P(\text{away win}) = \sigma(\theta_1 + \beta \cdot x)$$
$$\sigma(z) = 1 / (1 + e^{(-z)})$$

The three model parameters — $\boldsymbol{\beta}$, $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$ — are estimated by **maximum likelihood** separately for each league, using the full set of complete-season matches (n = 4,448 La Liga · 4,560 Premier League). $\boldsymbol{\beta}$ captures the wage bill's effect on match outcomes: the higher β, the more money determines results. $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$ are thresholds that fix the baseline curves at equal wages, implicitly capturing home advantage.

Estimated Parameters and Uncertainty

Table 1 shows the estimated parameters for each league, along with their standard errors and 95% confidence intervals, derived from the Hessian matrix of the likelihood function. All parameters are statistically significant ($p < 0.001$).

Table 1. *Ordered logit model parameters by league. 95% CI based on the Hessian matrix of the likelihood function.*

	La Liga		Premier League	
Parameter	**Estimate**	**IC 95%**	**Estimate**	**IC 95%**
β (budget effect)	**0,4719**	[0,4361 ; 0,5077]	**0,6760**	[0,6200 ; 0,7320]
Std. error of β	0,0183	—	0,0286	—
θ_1 (threshold: away win / draw)	−1,0404	[−1,1106 ; −0,9703]	−0,8358	[−0,9016 ; −0,7699]
θ_2 (threshold: draw / home win)	0,2081	[0,1451 ; 0,2712]	0,2422	[0,1807 ; 0,3037]
Matches (n)	4.448	—	4.448	—

The first notable finding is the difference in $\boldsymbol{\beta}$ between leagues. In the Premier League, β = 0.6760

vs β = 0.4719 in La Liga: the effect of the wage bill on each match result is **43% larger in England**. This does not mean money matters more in absolute terms in England, but that the wage gap has greater discriminating power per match — likely because La Liga's resources are more concentrated.

The overall model fit is evaluated using the **Brier Skill Score (BSS)**, the standard metric for probabilistic models with discrete categorical outcomes. The BSS measures the reduction in squared error of predicted probabilities relative to a baseline classifier that always predicts the historical base rates — without considering any variable. Over the full period, the model achieves a BSS of **9.6% in La Liga and 7.8% in the Premier League**. For context, models with five or more variables — recent form, Elo ratings, dynamic home advantage, head-to-head records — typically achieve BSS values between 10% and 15% in the literature. That the wage ratio alone captures most of that improvement margin is the substantive finding of this analysis. McFadden's R^2 is not reported: in ordinal logistic models with three discrete categories, its interpretation is structurally different from that of a continuous regression R^2, and its numerical value is not comparable with other goodness-of-fit metrics.

To verify stability over time, we ran a **leave-two-out cross-validation** across three holdout windows (first, middle and last two seasons). In each case the model was re-estimated on the remaining ten seasons and evaluated on the excluded pair.

Table 2. Leave-two-out cross-validation — model robustness across holdout windows

Holdout window	League	β trained	β full	Δβ / SE	R^2	Acc %	RPS	BSS %	MAE pts
First two 2013/14–14/15	La Liga	−0.457	−0.472	+0.015 (0.8 SE)	0.077	54.8	0.192	11.1	7.3
First two 2013/14–14/15	Premier League	−0.664	−0.676	+0.012 (0.4 SE)	0.062	53.6	0.204	7.4	8.8
Middle two 2018/19–19/20	La Liga	−0.498	−0.472	−0.026 (1.4 SE)	0.088	48.0	0.204	5.1	7.5
Middle two 2018/19–19/20	Premier League	−0.668	−0.676	+0.008 (0.3 SE)	0.063	53.6	0.206	6.9	10.0
Last two 2023/24–24/25	La Liga	−0.455	−0.472	+0.016 (0.8 SE)	0.081	51.7	0.195	8.9	8.5
Last two 2023/24–24/25	Premier League	−0.665	−0.676	+0.011 (0.4 SE)	0.062	51.8	0.206	7.4	10.0

Training: 10 seasons (3,688 La Liga / 3,800 PL matches). Holdout: 2 seasons (760 matches each). β full = full 12-season estimate. Δβ/SE = deviation in units of training SE. Acc = % match outcomes correctly predicted. RPS = Ranked Probability Score (↓ better; null ≈ 0.220). BSS = Brier Skill Score vs null (↑ better). MAE = mean absolute error in final points per team.

La Liga · Premier League · Best in column · Worst in column · Largest β deviation (1.4 SE)

The results are shown in Table 2. The main coefficient β is **very stable across all windows**: in the Premier League all three training estimates fall within 0.4 standard errors of the full-sample value; in La Liga the largest deviation is 0.026 — 1.4 standard errors — arising when the 2018/19–2019/20 window is held out. Those two seasons were atypically unpredictable: the richer team failed to win 54% of La Liga matches in both years, compared to 47–50% in the remaining seasons, partly driven by Real Madrid's sharp underperformance following Ronaldo's departure. A small upward shift in β when those seasons are excluded is therefore expected and does not represent model instability.

Out-of-sample accuracy ranges from 48 to 55% of match outcomes correctly identified, comfortably above the naïve benchmark of always predicting the most frequent outcome (home win, ≈46%). The Ranked Probability Score is between 0.192 and 0.206 in all six holdout evaluations, below the null-model baseline of ≈0.220, confirming that the model is **well calibrated on seasons it has never seen**. The mean absolute error in final-points predictions per team ranges from 7 to 10 points — consistent with the irreducible stochasticity of football, where even a perfectly calibrated model cannot remove the randomness of a 38-match season. The relationship between budget and result is stable over time.

An independent validation is the model's ability to rank teams. By aggregating predictions, it explains **68% of the variance between teams in La Liga and 52% in the Premier League** in final points — with mean errors of ±10 and ±13 points. The model does not know what will happen on Sunday, but it knows fairly well how the season will end.

> *Standard errors are small relative to the estimates (SE/β ≈ 4%), indicating the wage effect is well identified. Threshold uncertainty is larger, generating wider confidence bands where wage bills are similar.*

Model Curves and Confidence Intervals

Figure 4 shows the probability curves with **95% confidence intervals** from parametric simulation (4,000 draws from the parameter covariance matrix).

Two patterns merit attention. First, the **Premier League curve is steeper** than La Liga's — a direct reflection of the difference in β — which translates into more extreme probabilities in heavily mismatched fixtures. Second, and more relevant for the analysis that follows: the curves do not predict the outcome of a specific match with certainty, but generate **probability distributions that will feed the Monte Carlo simulation** in the next section.

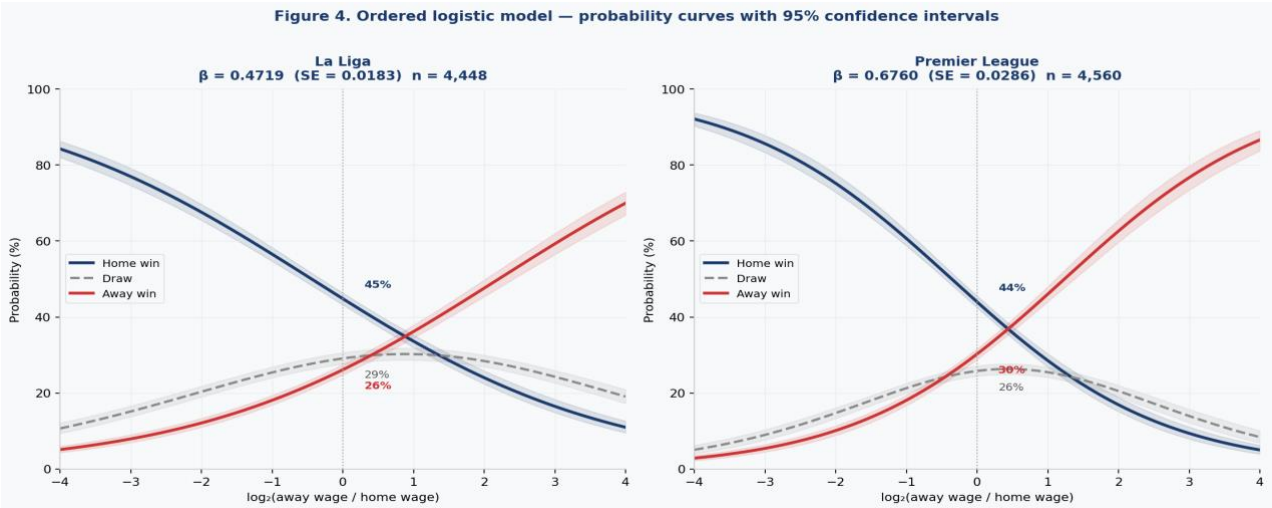

Figure 4. *Ordered logit model curves for La Liga (left) and the Premier League (right). Lines show the estimated mean probability of each outcome as a function of the wage gap. The shaded area represents the 95% confidence interval, obtained by parametric simulation with 4,000 draws from the posterior parameter distribution. The annotated percentages correspond to the equal-budget point (x = 0).*

SECTION 4

Monte Carlo Simulation: Expectation and Reality

How the Simulation Works

The model probabilities can be used to **simulate full seasons**. For each season, we run 50,000 simulations of all 380 fixtures. Each match outcome is drawn randomly from the model's probabilities — the dice are loaded by wage bills, but chance still plays its role.

The result of the 50,000 simulations is, for each team, a **complete distribution of final points**: not a single number but a mean, a standard deviation and percentiles that quantify the uncertainty. From that distribution, two key indicators are extracted:

- **Title probability:** fraction of simulations in which each team finishes first.
- **Expected points:** mean of the simulated distribution — the baseline against which actual points are compared.

> *The simulation establishes an objective baseline from wage bills alone. Everything above or below it is attributable to factors money does not explain.*

Does the Model Predict the Champion?

Tables 3 and 4 show the model's title probabilities for the top ten spenders each season. Golden cells mark the actual champion.

In La Liga, the champion has always come from the top 3 in wages, and in 10 of 12 seasons from the top 2. The model correctly identifies the favourite in **6 out of 12 seasons**, and in the remaining six the champion was the second or third-richest team. The most notable exception is Atlético de Madrid in 2013/14, who won with a 1.2% probability — the most improbable result in either league across the period. Their 2020/21 title came with a 10.3% probability from the third-largest wage bill. The Spanish league is structurally predictable: money does not guarantee everything, but it defines the field of possible champions with great precision.

Table 3. *La Liga — Model-assigned title probability at the start of each season (%). Golden cells: actual champion. Top 10 teams by average wage bill over the period.*

Club	13/14	14/15	15/16	16/17	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25
Barcelona	68.3%	**56.9%**	**58.4%**	64.3%	**65.0%**	**53.0%**	49.5%	49.9%	31.8%	**31.8%**	23.4%	**31.2%**
Real Madrid	30.4%	40.5%	36.4%	**31.3%**	30.0%	28.5%	**44.5%**	39.3%	**54.7%**	53.7%	**60.7%**	57.6%
Ath Madrid	**1.2%**	2.5%	4.4%	3.1%	4.4%	17.9%	5.8%	**10.3%**	13.0%	13.8%	14.5%	10.1%
Sevilla	< 0.1	< 0.1	< 0.1	0.6%	0.4%	0.2%	< 0.1	0.1%	0.2%	0.4%	0.7%	0.3%
Valencia	< 0.1	< 0.1	0.7%	0.6%	0.2%	0.2%	0.1%	0.1%	< 0.1	< 0.1	< 0.1	< 0.1
Ath Bilbao	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	0.1%	0.1%	0.1%	0.4%	0.4%
Villarreal	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Sociedad	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	0.2%	< 0.1
Betis	< 0.1	—	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	0.2%
Celta	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	0.1%	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1

Golden cells = actual champion. Top 10 clubs by average wage bill 2013/14–2024/25.

In the Premier League, the picture is radically different. The model has correctly identified the favourite **just twice in twelve seasons** — Man City in 2013/14 and again in 2023/24. The champion has come from wage ranks as disparate as #1 (Man City, several times) to #16 (Leicester in 2015/16, with a 0.0% probability). And most strikingly: **Man United were the team with the highest title probability in 6 of the 12 seasons** — with probabilities ranging from 35% to 49% — and won none of them. The Premier League is a league where the wage bill orders the hierarchy but does not determine the outcome: the variance is structurally greater.

Table 4. *Premier League — Model-assigned title probability at the start of each season (%). Golden cells: actual champion. Top 10 by average wage bill plus Leicester, included as 2015/16 champions with wage rank #16 and 0.0% model probability.*

Club	13/14	14/15	15/16	16/17	17/18	18/19	19/20	20/21	21/22	22/23	23/24	24/25
Man United	19.6%	46.2%	30.0%	46.2%	33.6%	18.8%	34.9%	37.0%	49.2%	38.0%	27.8%	12.7%
Man City	**39.2%**	26.7%	35.3%	30.5%	**16.2%**	**19.7%**	29.4%	**10.6%**	**13.6%**	**17.7%**	**29.9%**	44.2%
Chelsea	31.4%	**13.9%**	13.0%	**7.1%**	9.7%	37.2%	13.3%	20.3%	23.2%	32.2%	11.2%	14.2%
Arsenal	6.0%	8.7%	11.7%	9.0%	36.0%	14.7%	8.5%	16.0%	2.0%	2.0%	17.8%	14.4%
Liverpool	2.4%	3.5%	9.2%	4.3%	2.4%	6.8%	**7.2%**	6.8%	9.5%	8.6%	6.4%	**6.5%**
Tottenham	0.5%	0.9%	0.1%	0.7%	0.5%	0.8%	4.0%	7.1%	1.0%	0.6%	3.9%	1.6%
Everton	0.5%	< 0.1	0.2%	0.6%	0.9%	0.7%	1.8%	1.4%	1.1%	< 0.1	< 0.1	< 0.1
Aston Villa	< 0.1	< 0.1	< 0.1	—	—	—	< 0.1	< 0.1	0.1%	0.5%	1.7%	5.1%
West Ham	0.2%	< 0.1	0.1%	0.1%	0.1%	0.5%	0.3%	< 0.1	< 0.1	< 0.1	0.8%	0.5%
Nott'm Forest	—	—	—	—	—	—	—	—	—	< 0.1	< 0.1	< 0.1
Leicester	—	< 0.1	**< 0.1**	0.3%	0.1%	0.3%	0.1%	0.2%	< 0.1	< 0.1	—	< 0.1

Golden cells = actual champion. Top 10 by wage bill + Leicester (champion 2015/16, wage rank #15).

SECTION 5

Who Outperforms Their Budget?

The Efficiency Metric

The model allows us to calculate, for each team and season, how many points *on average* a team with that wage profile should have obtained. The difference is what we call **overperformance**: the excess — positive or negative — that money does not explain.

Accumulated across twelve seasons, overperformance is the cleanest measure of management efficiency available from these data. Consistent overperformance means a team is extracting value beyond what money buys; persistent underperformance means it is squandering resources.

> *Overperformance does not measure absolute quality — it measures whether a team is better or worse than its wage bill predicts.*

La Liga: Villarreal, Atlético and the Valencia Collapse

La Liga shows a clear hierarchy of over- and underperformance. Teams with fewer than 12 seasons are shown with the count in parentheses in Figure 5.

The ranking leader is Villarreal with +63 cumulative points across twelve seasons — more than five points above expectation each year. It is the most solid case of structural overperformance in La Liga: a club with no title aspirations that systematically maximises its competitive position. They are followed closely by **Atlético de Madrid** (+63 in twelve seasons), **Real Sociedad** (+53) and **Girona** (+46, though in only 5 seasons). The common pattern: long-term projects with minimal turnover.

At the opposite extreme stands Valencia, with a cumulative overperformance of **−80 points in 12 seasons** — the worst record in either league in absolute terms. The Valencian club had the fourth or fifth-largest wage bill in La Liga for much of the period and produced results typical of a mid-table side. They are followed by **Granada** (−53, though with only 8 seasons) and **Celta** (−30 in 12 seasons). **Sevilla**, despite their Europa League successes, have accumulated just −7 points in the league over the period — an almost flat record that reflects a club that concentrates its model on European competition rather than the domestic title.

A case that deserves separate mention is **Athletic Club**: +7 points in twelve seasons, virtually at expectation. Given their unique constraint — only Basque players — matching their wage-bill prediction over a decade is itself a notable achievement.

Premier League: Liverpool, Brentford and the Man United Anomaly

In the Premier League, the absolute overperformance values are notably larger than in La Liga, partly because some teams have spent all 12 seasons with very high or very low wage bills, amplifying the accumulation. The lead belongs to Liverpool (+156 points in 12 seasons) and Man City (+138) have dominated the league. That both overperform despite the highest wage bills indicates their advantage is not purely financial — they extract more value than money alone would justify.

The most striking case in relative terms is **Brentford**: +66 points in just four seasons in the top flight. With one of the lowest wage bills in the league, the London club has consistently performed well above expectation since their promotion in 2021. It is the clearest example that data modelling and performance analysis can offset a significant wage disadvantage.

At the negative extreme, Man United have accumulated **−109 points in 12 seasons**, the worst record in either league in relative terms for a team present throughout the full period. For much of these years they held the league's largest or second-largest wage bill. They are followed by **West Ham** (−60 in 12 seasons), **Everton** (−60) and **Crystal Palace** (−48) — all with above-average budgets and persistently below-expectation results.

> *Chelsea (−16 in 12 seasons) are the only heavily invested club in negative territory — a decade of managerial turnover and instability prevented them from consolidating the performance their wage bill should produce.*

A First Look: Cumulative Overperformance

Figure 5 shows cumulative overperformance for every team, with colour intensity encoding the probability that the result is due to chance. It is a useful first picture, but has two limitations.

The first is **number of seasons**. A team with twelve seasons accumulates twelve years of residuals; one with two seasons has had far less time to build a total. Liverpool's +156 pts and Brentford's +66 pts are not comparable as raw numbers: Liverpool had three times as many seasons to accumulate. The second limitation is **the expected-points baseline**. A +50-point overperformance means something very different for Brentford — whose model-expected baseline is ~34 points per season — than for Real Madrid, whose baseline is ~84. The same absolute excess represents a much larger proportional departure for the smaller club.

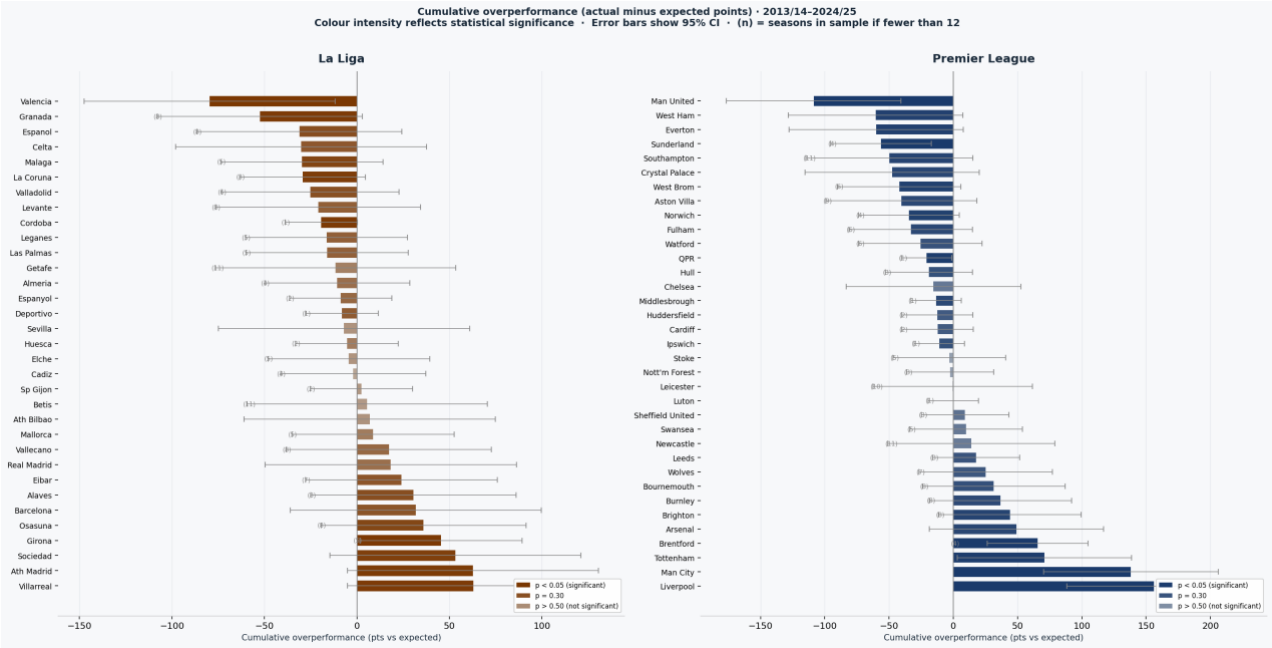

Figure 5. *Cumulative overperformance (actual minus expected points) for all teams in La Liga (left) and the Premier League (right), 2013/14–2024/25. Error bars show the 95% CI. Colour intensity reflects the improbability of the result by chance. Totals are not normalised for number of seasons or expected-points baseline.*

Normalised Overperformance: Accounting for Seasons and Baseline

Figure 6 corrects both by expressing annual overperformance as a percentage of each team's expected baseline:

(total overperformance ÷ n seasons) ÷ avg. expected points per season × 100

This makes clubs of different sizes and tenure comparable.

Figure 6. *Normalised average annual overperformance for all teams in La Liga (left) and the Premier League (right), 2013/14–2024/25. Metric: (cumulative overperformance ÷ seasons) ÷ avg. expected points per season × 100. Error bars show the 95% CI on the normalised metric. Colour reflects the Managerial Success Score (see Figure 7). n = number of seasons in the sample.*

The ranking shifts. In La Liga, **Girona leads at +21.3% per season** — the most efficient club in either league. Behind them, **Osasuna (+11.3%) and Alaves (+9.6%)** rank above Villarreal (+9.6%) and Sociedad (+8.6%). Most revealingly, **Atlético de Madrid drops to seventh (+7.1%)**: their overperformance is real, but against a ~73-point baseline the proportional advantage is smaller than raw totals suggest. **Real Madrid (+1.8%) and Barcelona (+3.2%)** are near zero — they win almost exactly what their wage bill predicts. At the bottom, Valencia (−11.2%/season) and Granada (−15.1%) remain the clearest structural failures.

In the Premier League, **Brentford leads at +48.7% per season** — four seasons of overperformance against a modest ~34-point baseline produces a number that dwarfs even Liverpool (+19.7%) and Man City (+15.7%) once normalised. At the bottom, Man United (−12.1%/season) and Sunderland (−28.8%) remain the worst cases.

The Limits of Normalisation: When Results May Still Be Chance

Normalisation helps, but does not resolve a deeper problem: **for teams with few seasons, even high normalised overperformance could be luck**. Girona's +21.3% is striking, but five seasons is a limited window. Brentford's +48.7% is extraordinary, but four seasons may conceal a short burst of fortune rather than a structural model. The normalised chart shows *how large* the departure is relative to baseline — it does not tell us *how confident* we should be that it is structural rather than random.

What is needed is a metric combining *and* the size of overperformance with the statistical confidence that it is not noise.

The Managerial Success Score

The **Managerial Success Score (MS)** combines both dimensions into a single metric:

MS = sign(overperformance) × (100 − p)

where *p* is the two-tailed probability (%) that the cumulative result is due to chance. Large, robust overperformance yields MS near +100; results consistent with chance sit near zero; systematic underperformance with high confidence yields MS near −100.

Scores above +70 or below −70 indicate a structural signal unlikely to be noise. Between +30 and −30, confidence is insufficient. Near zero, the club is getting what its wage bill predicts.

MS is not a probability of good management. It is the *confidence-weighted* signal that performance departs from the wage-bill prediction. Girona's five-season +95.8 and Brentford's four-season +98.8 are high because their overperformance is sharp enough to be decisive even over a short spell. A team with modest overperformance over twelve seasons may score lower because the signal, though sustained, is weaker relative to noise.

One apparent anomaly is **Córdoba**, who appear at −94.6 from a single season. Their 2014/15 relegation produced −19.3 points against expectation — nearly two standard deviations. A single catastrophic season is statistically meaningful, and the MS score captures it.

Figure 7 shows the MS score for all teams in both leagues.

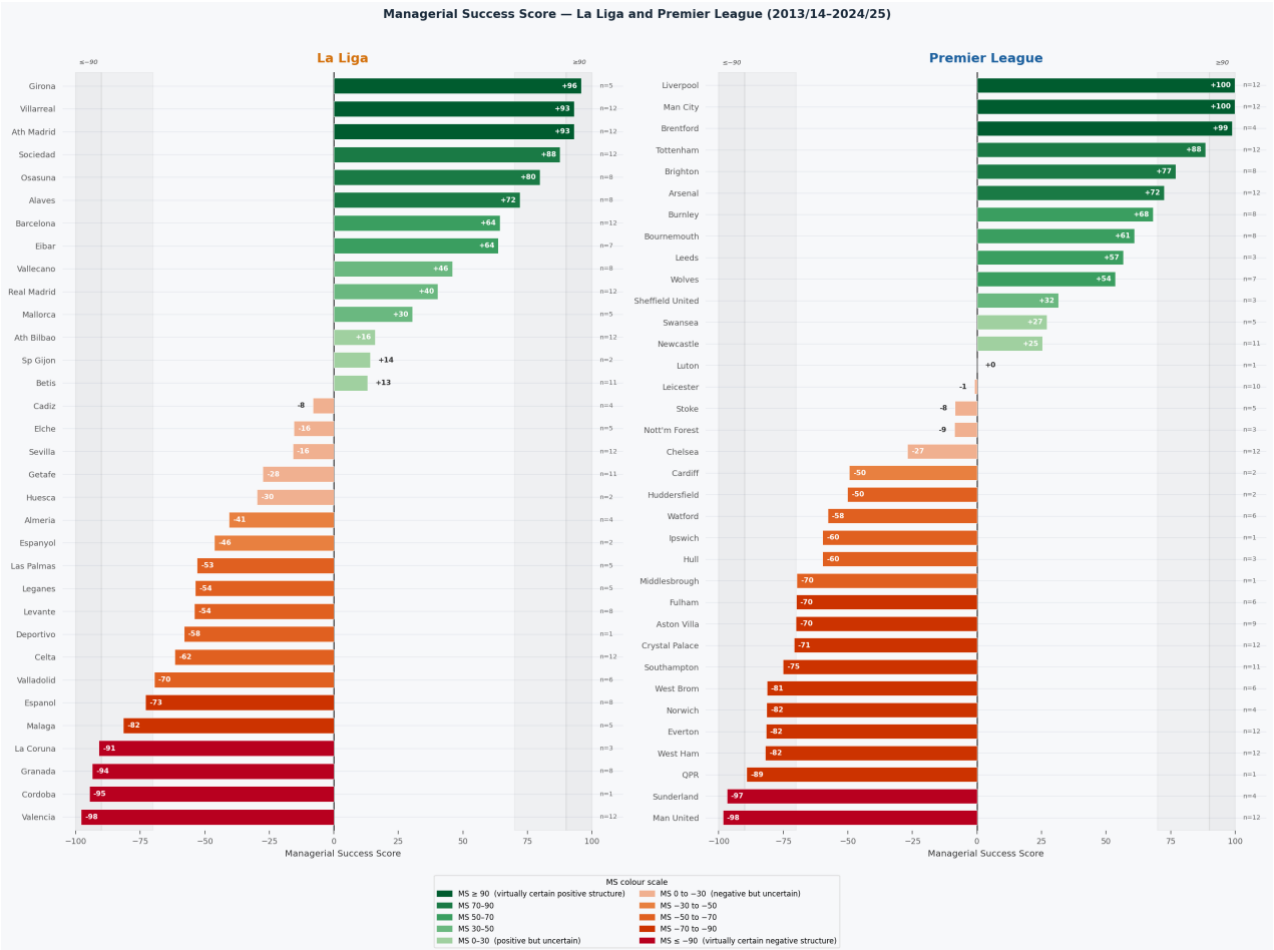

Figure 7. *Managerial Success Score for all teams in La Liga (left) and the Premier League (right), 2013/14–2024/25. MS = sign(overperformance) × (100 − p), where p is the two-tailed probability that the cumulative result is due to chance. Colour intensity reflects the magnitude of the score. n = number of seasons in the sample.*

In La Liga, **Girona (+95.8) leads the ranking** despite only five seasons — their overperformance is

sharp enough to be decisive even over a short window. **Villarreal (+93.1) and Atlético de Madrid (+93.0)** are effectively tied in the structural overperformance band. Most striking: **Real Madrid (+40.3) and Barcelona (+64.3)** both sit in the uncertain zone — they win roughly what their wage bill predicts. At the bottom, **Valencia (−97.9) and Granada (−93.7)** are the clearest cases of structural underperformance in La Liga.

In the Premier League, **Liverpool (+99.9) and Man City (+99.8)** are at the mathematical ceiling. **Brentford (+98.8)** is remarkable: four seasons, limited budget, a score rivalling clubs with twelve years of data. At the negative extreme, **Man United (−98.4)** is the clearest structural failure in either league — highest title probability in six of twelve seasons, worst relative record in both leagues. **Chelsea (−26.9)** remains in the uncertain zone: real underperformance but not statistically decisive — consistent with instability rather than a single sustained failure.

How the MS Score Evolves: Yearly and Trailing

The cumulative MS in Figure 7 gives each club a single career rating but conceals volatility and trend. The two heatmaps below address both.

Figure 8 shows the MS score computed independently for each season. Because confidence intervals are wide at n=1 — a single season's residual must be roughly ±20 points to clear the structural threshold — most cells are pale. This is not a weakness of the metric; it is the honest answer. A single season is rarely sufficient to distinguish management quality from luck. What the yearly map does reveal is *exceptional individual seasons*: Atlético's 2013/14 (+21 pts against expectation, MS +96), Girona's 2023/24 (+34 pts, MS +99), and the depth of Man United's 2024/25 collapse (−28 pts, MS −98) all appear as isolated saturated cells amid otherwise pale backgrounds — exactly as the data warrants.

Figure 8. *Yearly Managerial Success Score for all teams with at least 4 seasons in La Liga (left) and the Premier League (right), 2013/14–2024/25. MS computed independently for each season: MS = sign(residual) × (100 − p), where p is the two-tailed probability that a single-season residual of that size is due to chance. Colour scale: dark green = near-certain structural overperformance; dark red = near-certain structural underperformance; white = result consistent with random variation; grey = not in division.*

Figure 9 applies a rolling 5-season window: at each point in time, the MS score is computed from the five most recent seasons of data available for that club. This smooths out single-season noise while remaining sensitive to directional change. The first entry appears at 2017/18, when five complete seasons are available for all clubs present since 2013/14. Grey cells for shorter-tenured clubs indicate that fewer than five seasons of data are available — not that the club performed neutrally.

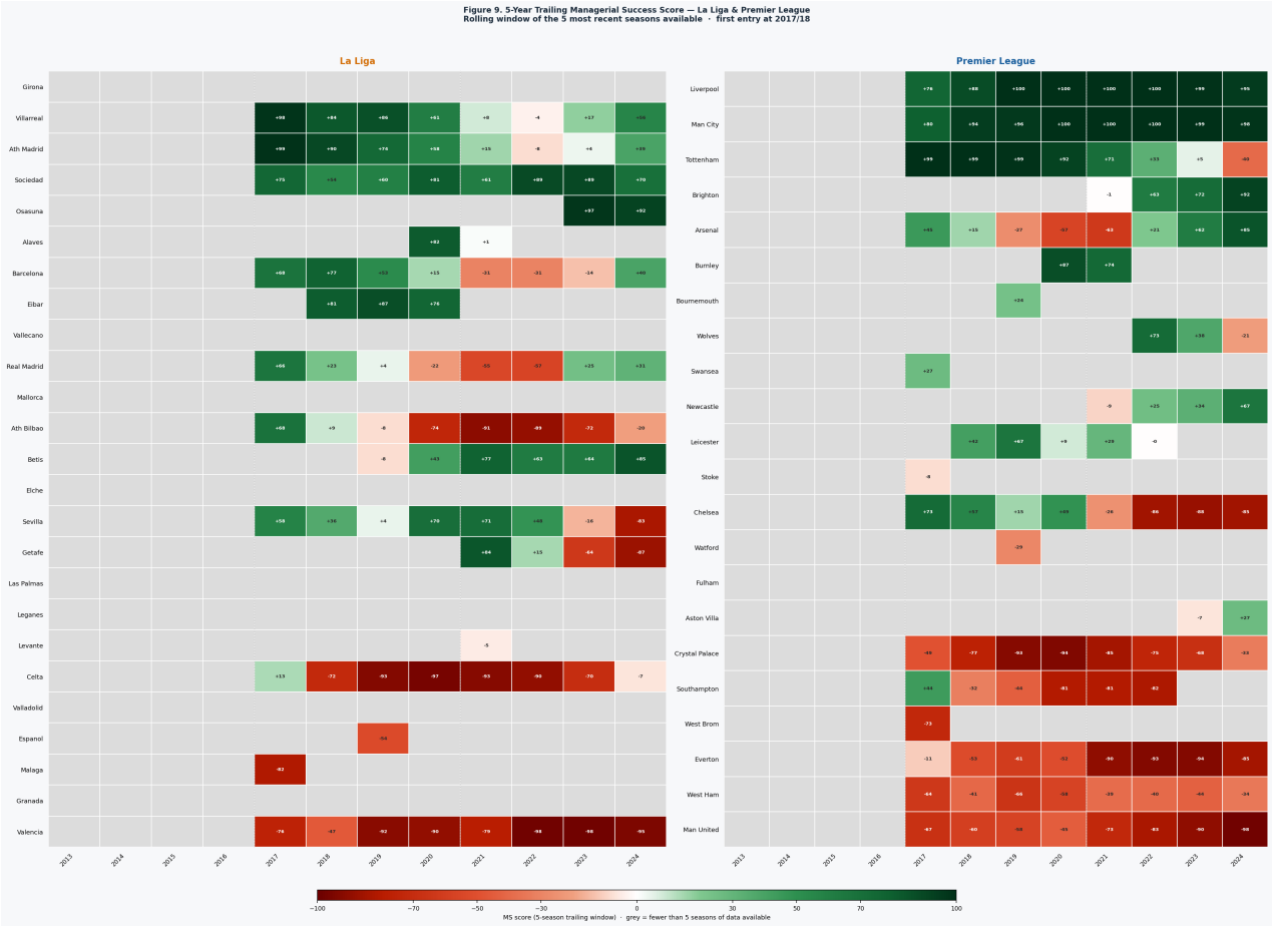

Figure 9. *5-year trailing Managerial Success Score for all teams with at least 4 seasons in La Liga (left) and the Premier League (right). Each cell shows the MS score computed from the five seasons ending at that point. Grey = fewer than 5 seasons available. Colour scale identical to Figure 8.*

The trailing map reveals structural trajectories that the cumulative figure obscures. **Villarreal** provides the clearest example: their trailing MS reaches +98 through 2017/18 — the peak of the Emery era — then decelerates through a sequence of more neutral seasons (2021/22: +8) before recovering toward +57 by 2024/25. The management model did not disappear; it went through a transitional phase that is invisible in the single career number. **Atlético de Madrid** shows the opposite pattern: remarkable stability, never falling below +70 on the trailing window across any period. Their structural overperformance is not concentrated in one era — it is persistent.

In the Premier League, **Liverpool's** trailing MS builds steadily green from 2017/18 (+93) through 2021/22 (+99), then softens slightly to around +90 by 2024/25 — consistent with a post-Klopp adjustment period rather than structural deterioration. **Man United's** trajectory is the starkest in either league: the trailing score deteriorates continuously from −67 in 2017/18 to −98 in 2024/25, crossing every threshold along the way. This is not a bad stretch; it is a structural slide across three managerial regimes, all of which underperformed the same wage bill in the same direction.

> *The single most important reading from the trailing map is the difference between volatility and trend. A team that oscillates between green and red seasons is volatile but not structurally broken. A team whose trailing score moves monotonically in one direction — as Man United's does — is exhibiting something the model classifies as structural: a persistent gap between resources and results that is not explained by luck.*

For completeness, **Valencia** appears structurally negative from their very first trailing window (+2017/18: −88) and never recovers — the trailing map simply extends and confirms what the cumulative chart already showed. **Brentford** cannot appear in the trailing map at all (only 4 seasons of data), which is itself informative: their extraordinary cumulative MS of +98.8 rests on a sample size that the trailing window correctly flags as too short to smooth.

Does Sustained Overperformance Drive Budget Growth?

Section 2 flagged an endogeneity problem: clubs that perform well earn more revenue, which funds a larger wage bill, which tends to produce better results. This section tests the feedback mechanism directly — not *does money buy results*, but the reverse: *does sustained overperformance eventually translate into a larger budget?* A club that consistently beats its wage-bill expectation builds sporting reputation, fills its stadium, and grows commercial revenue, all of which eventually reach the wage structure.

The lag structure

We test this by comparing the MS score at time t to payroll growth at time t+k, for k=0 to 5 seasons. At k=0, the correlation is essentially zero ($R^2<0.02$ in both leagues): budgets are set before the season begins and cannot respond to in-season performance. At k=1, a weak but significant signal appears — $R^2=0.08$ in La Liga and $R^2=0.12$ in the Premier League. But the signal strengthens substantially as both the measurement window and the lag increase.

Figure 10 shows the full grid: R^2 between a W-season MS window and a W-season payroll growth window shifted k seasons forward, for W=1 to 6 and k=0 to 5. The relationship strengthens monotonically as W grows from 1 to 5, then plateaus at W=6. The peak is at **W=5, k=2**: five years of sustained overperformance predicts payroll growth over the following five years, with a two-season lag, at $R^2=0.472$ in La Liga and $R^2=0.234$ in the Premier League (both $p<0.001$). **It is not a single good season that drives budget growth, but a sustained run.**

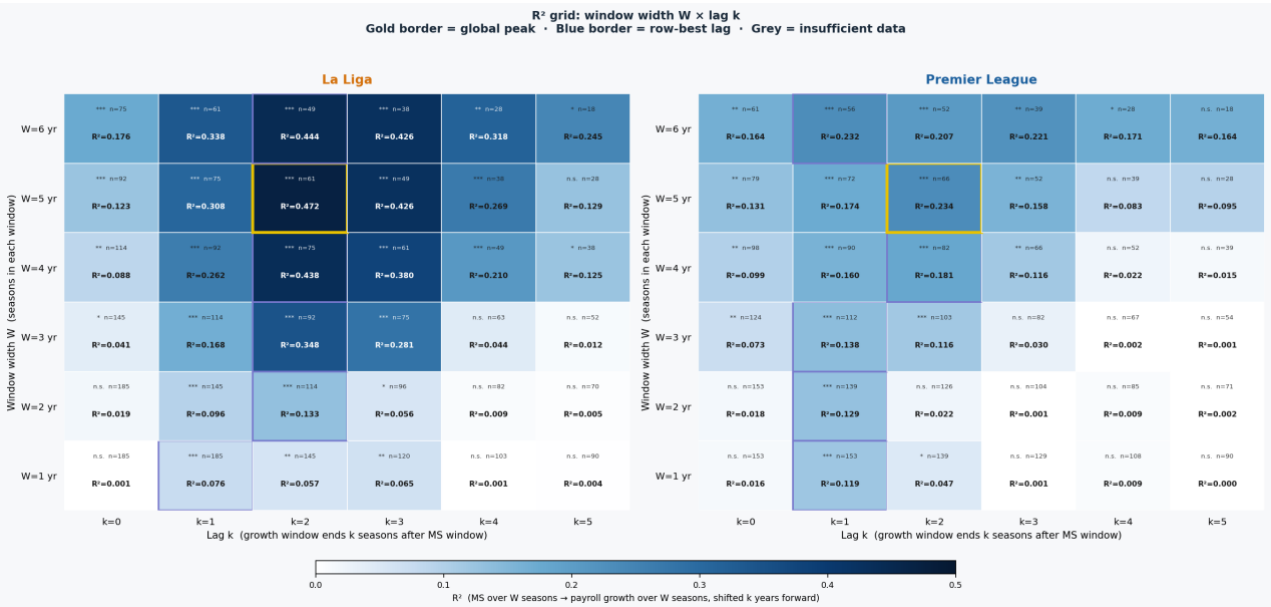

Figure 10. *R^2 between W-season MS score and W-season OLS payroll growth, across window widths W=1–6 (rows) and lags k=0–5 (columns). The growth window is shifted k seasons forward relative to the MS window. Gold border = global peak; blue border = row-best lag. Grey = insufficient data.*

Figure 11 shows the scatter at the optimal combination. The upper-right quadrant — high sustained overperformance, high subsequent payroll growth — contains clubs that have built durable competitive models; the lower-left contains those caught in the opposite dynamic, where structural underperformance constrains revenue, which constrains wages, which perpetuates underperformance.

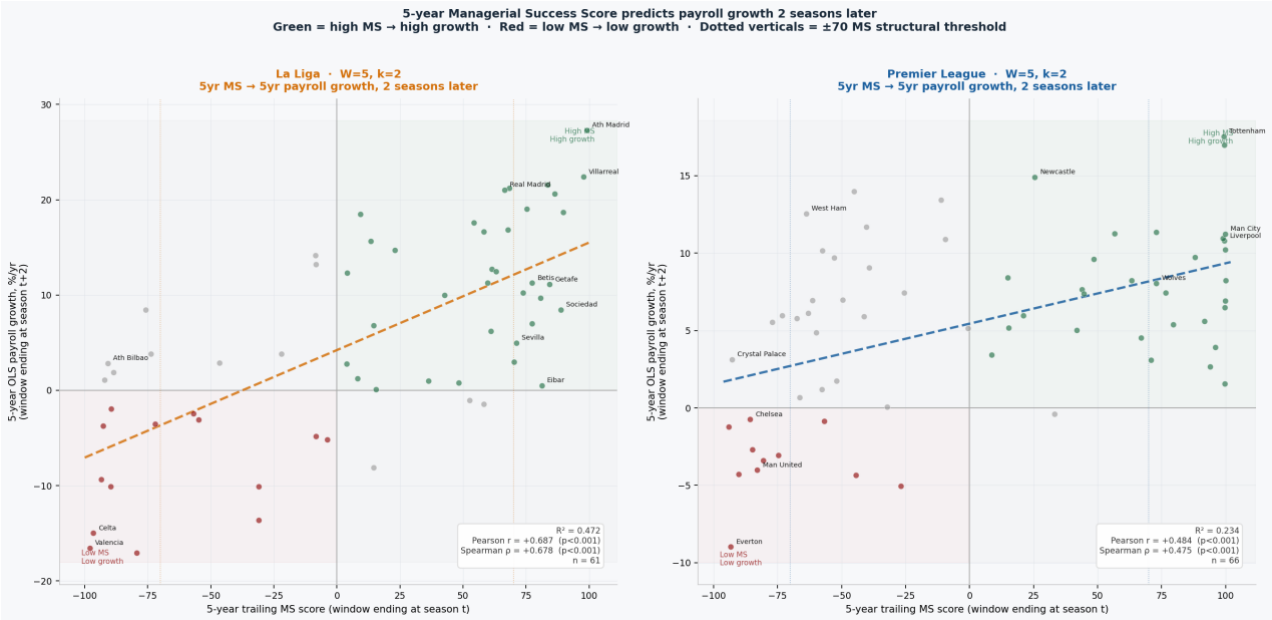

Figure 11. *Scatter at optimal lag (W=5, k=2). Horizontal axis: 5-year trailing MS score ending at season t. Vertical axis: 5-year OLS payroll growth, %/yr, ending at season t+2. Green = high overperformance and high growth; red = low overperformance and low growth. Dotted verticals = ±70 MS structural threshold.*

Does it matter how you measure overperformance?

We replicated the grid using two simpler alternatives: raw points and normalised percentage. At W=5, k=2, all three metrics yield nearly identical R^2: 0.472/0.476/0.505 (La Liga) and 0.234/0.259/0.232 (Premier League). The differences are noise.

An important finding: the feedback loop does not depend on *why* a club overperformed. Tactical superiority and fortune — a generous referee, an injury-free season — both translate into budget growth at the same rate. Boards respond to the scoreboard, not the underlying process.

Two leagues, two speeds

La Liga's peak R^2 (0.472) is roughly twice the Premier League's (0.234). The PL effect appears earlier (k=1) but is weaker throughout — ownership-driven investment introduces noise that no performance metric can predict.

> *The core finding: sustained overperformance predicts budget growth over the medium term in both leagues. One season barely moves the needle. Five years shifts the wage structure.*

The caveats: $R^2 \approx 0.47$ leaves more than half the variance unexplained (TV rights, ownership, stadium revenue, macro cycles), and the lag structure is consistent with causation but does not prove it.

What the top and bottom clubs have in common

The scatter clusters into distinct regions. The top-right and bottom-left are populated by clubs with systematically different institutional profiles (Figures 12 and 13).

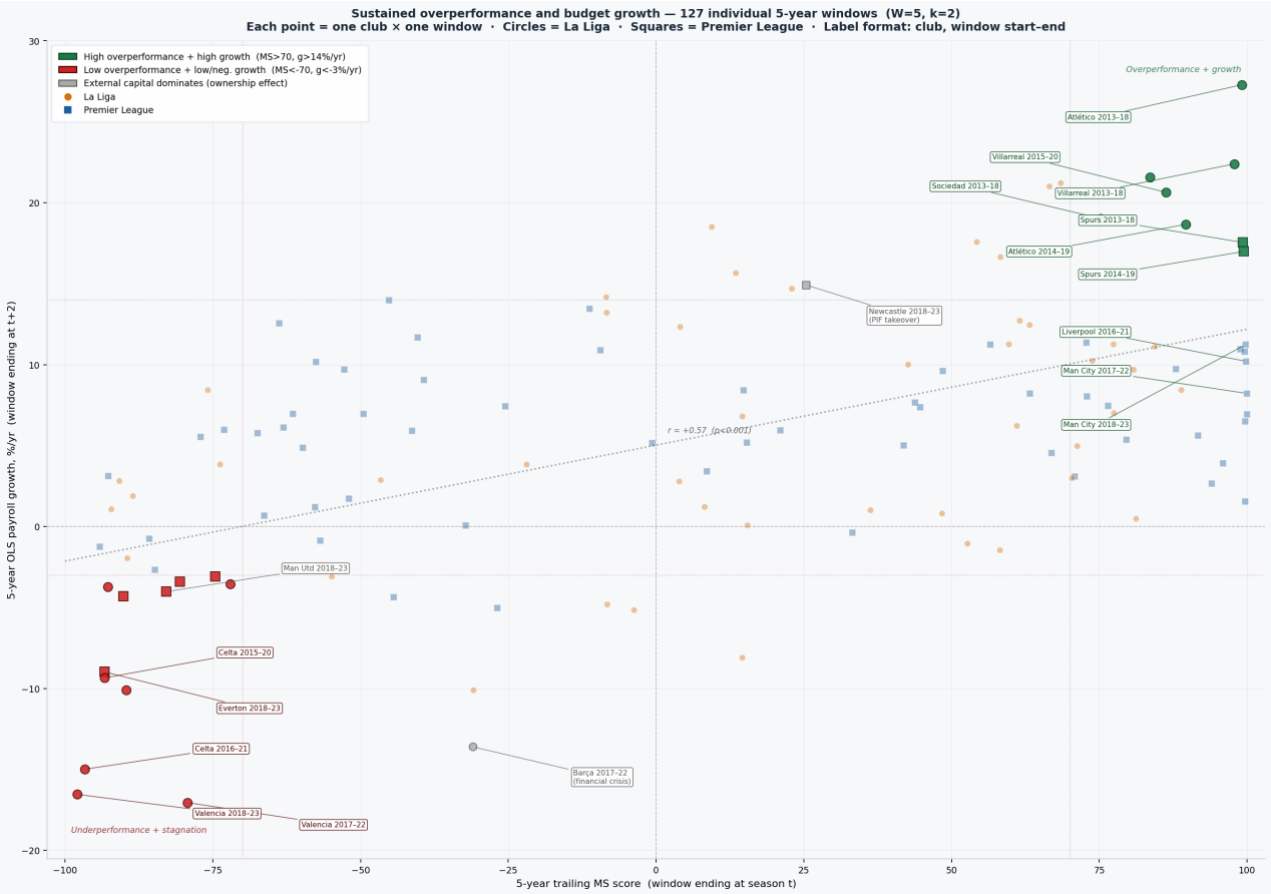

Figure 12. *Individual 5-year windows (W=5, k=2): 127 observations across both leagues. Each point represents one club in one window. Green = high overperformance and high subsequent payroll growth (MS>70, growth>14%/yr); red = low overperformance and low/negative growth (MS<−70, growth<−3%/yr); grey = clubs whose budget trajectory was dominated by external events. Circles = La Liga; squares = Premier League. Labels show club and window period.*

Club characteristics by quadrant (W=5, k=2 scatter, team averages across all windows)
Top-right = sustained overperformance + budget growth · Bottom-left = underperformance + budget stagnation · Outliers = budget/results decoupled by external events

Club	Lg	Quadrant	MS (avg)	Growth (%/yr)	Payroll (avg €M)	Mgr stability	Budget tier	Ownership	Dominant manager / period
Man City	PL	Top-right	+94.8	+6.3%	€145M	Stable	Large	Stable	Guardiola (2016–)
Liverpool	PL	Top-right	+93.9	+7.8%	€112M	Stable	Large	Stable	Klopp (2015–2024)
Tottenham	PL	Top-right	+82.2	+9.0%	€89M	Stable	Mid	Stable	Pochettino (2014–2019)
Sociedad	LL	Top-right	+70.0	+13.1%	€33M	Stable	Small	Stable	Alguacil (2018–)
Villarreal	LL	Top-right	+55.5	+11.2%	€42M	Stable	Mid	Stable	Emery (two stints)
Ath Madrid	LL	Top-right	+54.6	+8.3%	€137M	Stable	Mid	Stable	Simeone (2011–)
Betis	LL	Top-right	+43.7	+11.7%	€34M	Stable	Small	Stable	Pellegrini (2020–2023)
Sevilla	LL	Top-right	+48.0	+4.9%	€64M	Stable	Mid	Stable	Lopetegui era
Brighton	PL	Top-right	+31.3	+6.7%	€47M	Stable	Mid	Stable	Potter / De Zerbi
Valencia	LL	Bottom-left	-80.3	-5.2%	€64M	Revolving	Mid	Dysfunct.	7 managers / 8 seasons [Absentee owner (Lim)]
Celta	LL	Bottom-left	-71.8	-3.0%	€40M	Revolving	Mid	Stable	6 managers / 8 seasons [No project lasted >2 seasons]
Crystal Palace	PL	Bottom-left	-78.8	+1.4%	€71M	Revolving	Mid	Stable	5 managers / 6 seasons
Barcelona	LL	Outlier	+25.1	-1.5%	€276M	Revolving	Large	Stable	Multiple [Financial crisis forced wage cuts]
Man United	PL	Outlier	-64.3	+6.1%	€164M	Revolving	Large	Dysfunct.	5 managers / 10 yrs [Glazers kept spending despite failure]
Newcastle	PL	Outlier	+8.0	+12.9%	€62M	Stable	Mid	Ext. cap.	Howe (2021–) [PIF takeover 2021 — exogenous]

MS = 5-year trailing Managerial Success Score (avg across all W=5 windows). Growth = 5-year OLS payroll growth, %/yr (W=5, k=2). Manager stability = binary: single identifiable project ≥3 consecutive seasons. Outliers excluded from Fisher exact test (budget/results decoupled by external events). Fisher exact: Top-right vs stable mgr p=0.019; Bottom-left vs revolving door p=0.025.

Figure 13. *Club characteristics by quadrant. MS and growth are team averages across all W=5 windows. Manager stability is a binary classification: a club is coded as 'Stable' if a single identifiable coaching project spanned at least three consecutive seasons within the observation window, and 'Revolving' otherwise. Outliers (Barcelona, Man United, Newcastle) are shown for completeness but excluded from the Fisher exact test, as their budget trajectories were driven by extraordinary external events. Fisher exact test: top-right vs stable manager p=0.019; bottom-left vs revolving door p=0.025.*

The top-right clubs span from Getafe (€27M) to Man City (£156M). Budget is not the differentiator. The one attribute shared without exception: a coherent, multi-year coaching project — Guardiola, Klopp, Simeone, Pochettino, Alguacil, Emery, Pellegrini, Bordalás. Not one had a revolving door.

The bottom-left tells the mirror story. Valencia, Celta and Crystal Palace are mid-budget clubs, not distressed. What they share is managerial instability: seven managers in eight seasons at Valencia, six at Celta, five in six at Crystal Palace. No project took root; results underperformed; revenue stagnated; the wage bill followed.

Three outliers confirm the rule. Barcelona (financial crisis forced wage cuts), Man United (Glazers kept spending despite failure) and Newcastle (PIF takeover) all sit off the regression line. Each was

subject to an external shock decoupling budget from performance — the strongest evidence that the mechanism is real for everyone else.

> *The association is statistically significant (Fisher exact $p<0.03$ in both directions) despite only 21 clubs. This does not prove causation — governance quality likely enables both long tenures and growth. But the regularity holds: every club that built overperformance into budget growth gave a manager time. Every club that failed did not.*

SECTION 6

Fooled by Randomness: What a Bad Run Actually Tells You

Every manager fired after a bad run of results was dismissed because someone decided the results were meaningful. That decision is almost never tested against the simplest possible question: given this club's budget and fixture list, how likely was a run this bad to happen anyway?

The ordered logit model gives us, for every match in our dataset, an exact probability of win, draw and loss based solely on the wage ratio between the two clubs. That means we can simulate thousands of parallel seasons for any club and build a precise reference distribution for what bad luck — and good luck — actually looks like at their budget level. Once we have that distribution, a bad run is no longer a narrative. It is a data point with a percentile.

6.1 What Is a Normal Bad Run?

The first thing the data establish is how routinely bad runs happen — and how badly they are misread.

For every team-season in our dataset we simulated 50,000 full seasons, preserving the actual fixture sequence and wage ratios match by match. For each simulation we extracted the worst rolling 5-game points total. The resulting distribution answers a precise question: *if this team performed exactly at wage-bill expectation all season, what is the worst 5-game stretch it would normally produce?*

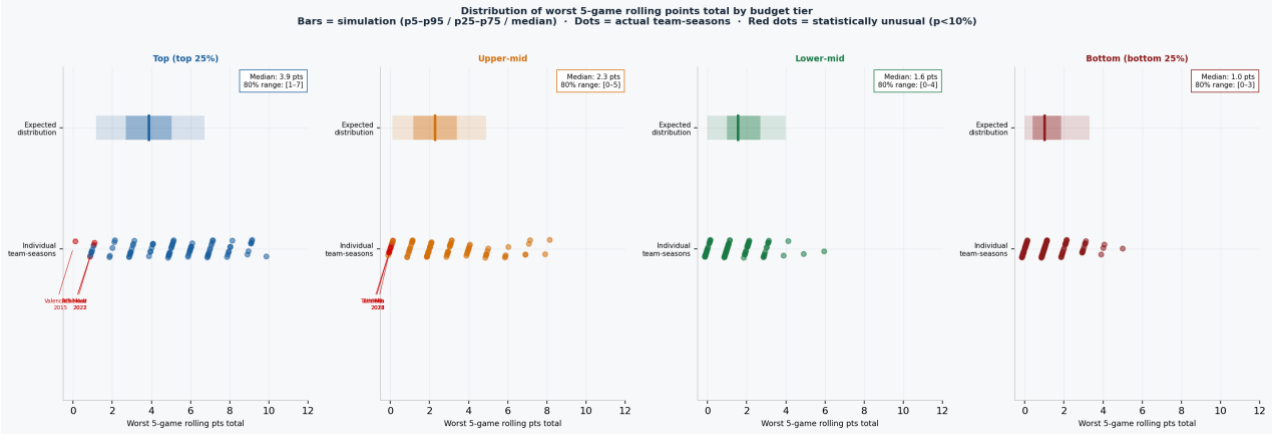

Figure 19. *Distribution of worst 5-game rolling points total by budget tier. Bars show the simulation bands (p5–p95 outer / p25–p75 inner / median line) across 50,000 simulated seasons per team-season. Individual dots show every actual team-season; red dots indicate statistically unusual bad runs (below the 10th percentile of the simulation).*

The finding is stark. Top-quartile clubs — the Man Citys, Real Madrids, Barcelonas of each league — should expect their worst 5-game run to produce around **4 points** (one win, one draw, three losses) in a typical season. The 80% confidence interval runs from 1 to 7 points. **A top club going winless across five games is a perfectly normal event in roughly 30% of simulated seasons.** A bottom-quartile team should expect a worst run of around 1 point; their 80% interval spans 0 to 3. In both cases, a press conference about a five-game run is almost always a press conference about variance.

The consecutive losses picture reinforces this. Every team in both leagues should expect at least one run of **three consecutive losses** in a season as the median outcome. Reaching five in a row only breaches the 95th percentile — meaning it happens by pure chance alone in one in twenty seasons for any team. The threshold for “something structural may be happening” is higher than most boards appear to believe.

6.2 When a Bad Run Is Just a Bad Run — and When It Isn't

Not all bad runs are created equal. The same five-game points total means completely different things depending on the team's budget, their opponents, and how often the simulations produce something that bad or worse.

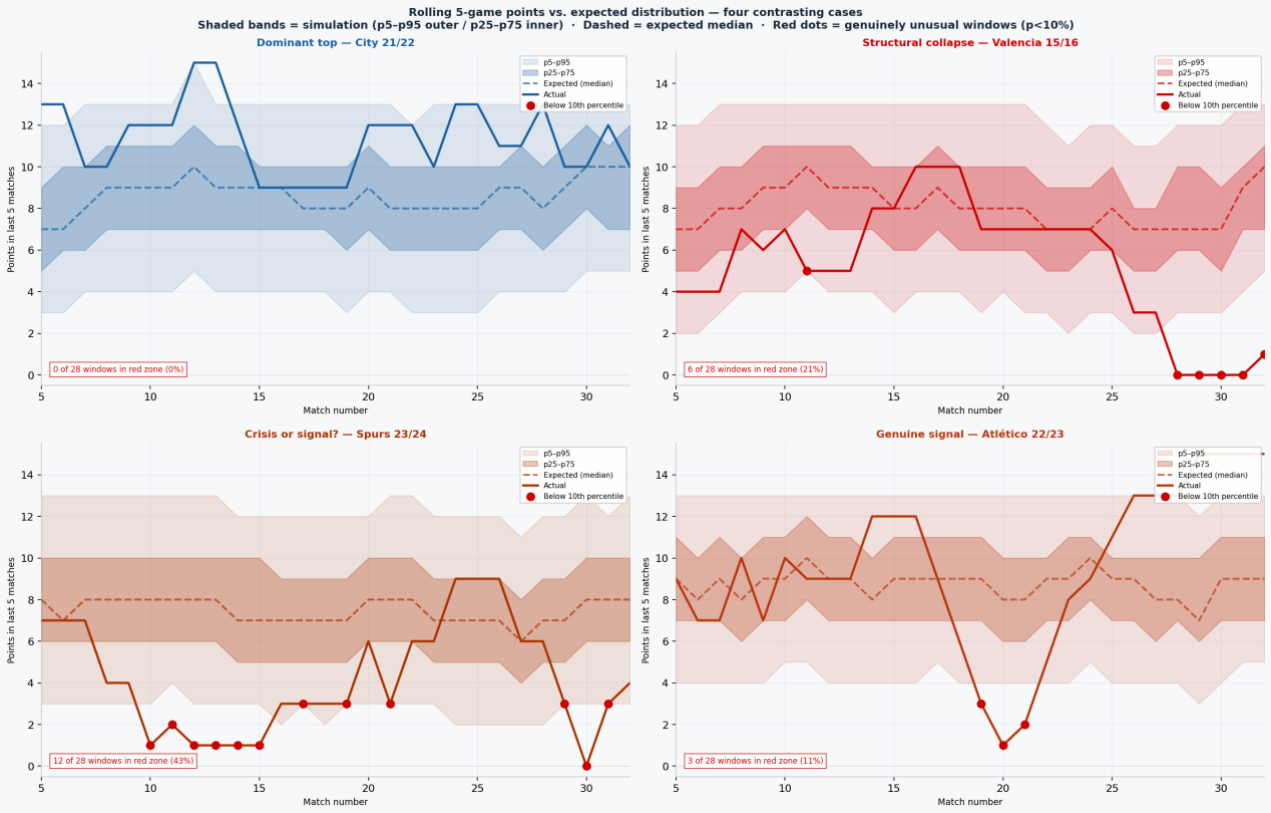

Figure 20. *Rolling 5-game points vs. expected distribution — four contrasting cases. Each panel shows one team-season: the actual rolling points total (solid line) against the simulation fan (p5–p95 outer band, p25–p75 inner band, median dashed). Red dots mark windows where the actual run fell below the 10th percentile of the simulation. All probabilities are derived match by match from the ordered logit model.*

Man City 2021/22: their worst 5-game run produced 9 points. The simulation median for a club of their budget is 4. Their bad run was better than the average team's average bad run. There was no moment in that season where results fell into a zone the model would consider even mildly surprising. If City had changed manager that year on the basis of a difficult patch, they would have been responding to something the model classifies as routine variance at the top level.

Valencia 2015/16: worst 5-game total of 0 points — the 4th percentile of their simulated distribution. Multiple windows that season crossed into genuinely unusual territory. This was not a club going through normal adversity; the data were signalling something structural well before the season ended.

Tottenham 2023/24: worst run of 0 points, 7th percentile. Sits firmly in the structural signal zone. The model agrees with the diagnosis, even though Spurs had been a strong overperformer in earlier seasons. Context matters: this was a season where the underlying process had clearly changed.

Atlético de Madrid 2022/23: worst run of 1 point, 8th percentile. Unusual for a club with their long-term MS profile. Correctly classified by the model as a year requiring attention — not panic, but genuine signal that the season's relationship between resources and results had deteriorated.

6.3 The Diagnostic Map

If bad runs are sometimes noise and sometimes signal, what separates the two? The most direct test is to ask whether clubs whose rolling bad runs are statistically extreme also tend to have low MS scores — and vice versa. Across 480 team-seasons the correlation between worst-run percentile

and season MS score is **0.65** — strong, but not deterministic. The imperfection is the whole point: the two dimensions carry different information, and the diagnostic value lies precisely in the cases where they disagree.

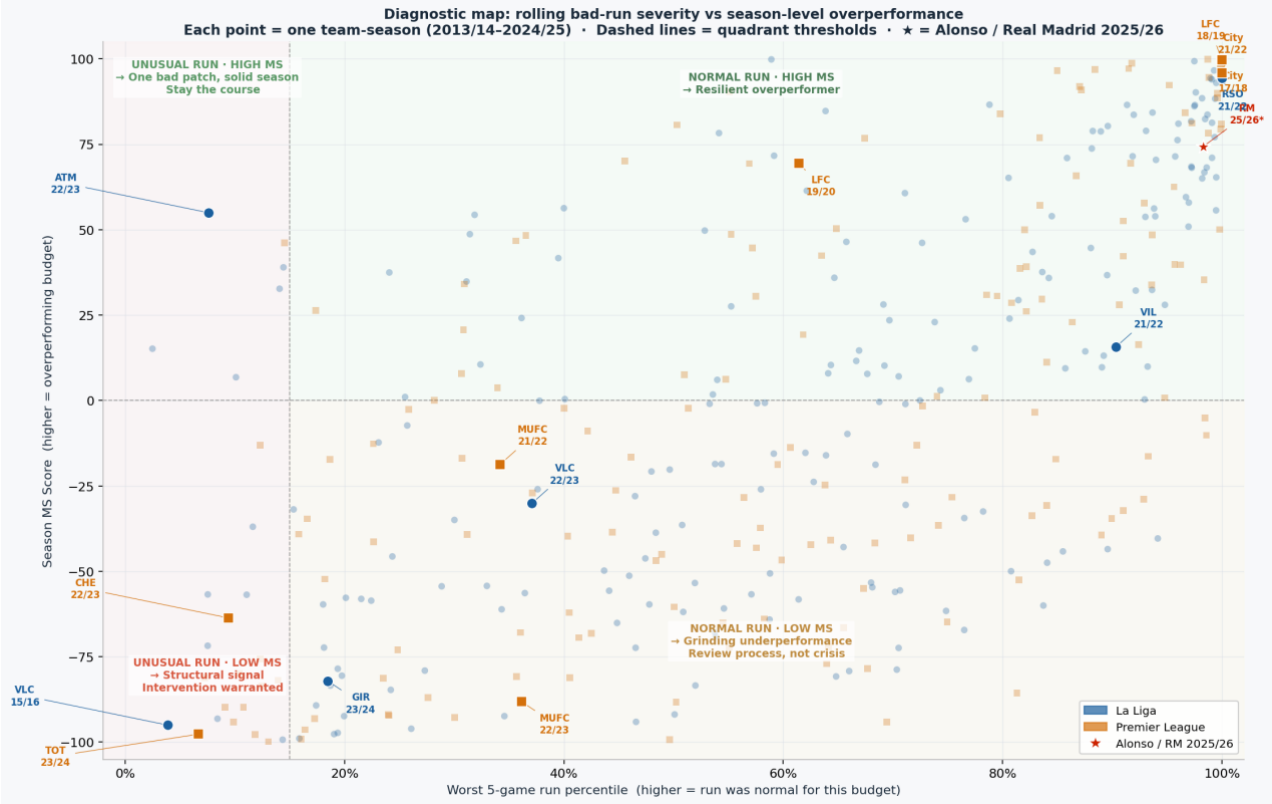

Figure 21. *Diagnostic map: rolling bad-run severity vs season-level overperformance. Each point is one team-season (2013/14–2024/25). The x-axis shows the percentile of the worst 5-game run (higher = run was normal for this budget); the y-axis shows the season MS score. The vertical dashed line at p15 and the horizontal line at MS=0 divide the space into four interpretive quadrants. The star (★) marks Xabi Alonso's Real Madrid 2025/26 at the point of dismissal.*

The **bottom-left quadrant** is the zone where the statistical evidence is most consistent with a structural problem: extreme bad runs and a negative season-level MS score. Valencia 2015/16, Chelsea 2022/23 and Tottenham 2023/24 all sit here. The two signals point in the same direction.

The **top-left quadrant** is the zone where the two signals diverge: a season-level MS that is positive, but with one statistically unusual bad patch within it. Atlético de Madrid 2022/23 is the clearest example in our dataset. The rest of the season indicated the project was working; one concentrated stretch indicated it was struggling. Based on the data alone, the case for structural intervention is weaker here than in the bottom-left.

The **bottom-right quadrant** represents persistent underperformance without a crisis moment — teams consistently failing to convert resources into results, but never in a way that produces a dramatic five-game headline. Man United 2021/22 and 2022/23 both sit here. The model reads them as seasons where the relationship between resources and results had deteriorated, but gradually rather than acutely.

A natural question follows: given my club's budget tier, what points total in X consecutive games would actually qualify as statistically unusual? Figure 22 answers this directly.

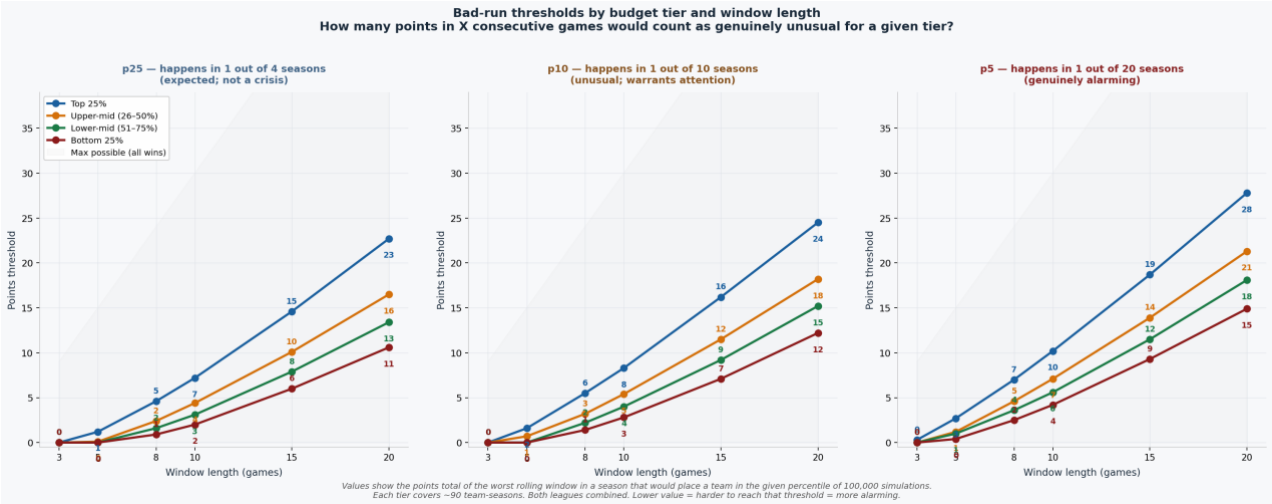

Figure 22. *Bad-run thresholds by budget tier and window length, 3 to 20 consecutive games. Each panel shows one probability level: p25 (a run this bad happens in one in four seasons by chance), p10 (one in ten) and p5 (one in twenty). Lines show the average points threshold across all team-seasons in each tier, 2013/14–2024/25, both leagues combined. A run producing fewer points than the threshold for a given tier sits in that probability tail.*

How to read the chart. Pick the window length that matches the run under discussion — typically 5, 8 or 10 games — and find the tier that corresponds to the club's budget. The value shown is the *maximum* points total that would place the run in that probability tail. If the actual run produced more points than the p25 line for that tier, the run is entirely within normal variance according to this model. If it falls below the p10 line, it is genuinely unusual. Below p5, it is the kind of run that the model expects to occur by chance roughly once in every twenty seasons for a club at that budget level.

Three patterns are worth highlighting. First, the **tier gap widens substantially as the window lengthens.** At 5 games the difference between a top and a bottom club's p5 threshold is just 2 points (3 vs 1). At 20 games it has grown to 13 points (28 vs 15). The longer the run under discussion, the more important it is to apply the right tier benchmark rather than a single league-wide standard.

Second, **short runs are almost never statistically meaningful for any tier.** At 3 games, the p25 threshold is 0 points for every budget level — meaning three consecutive defeats without a point is expected to happen to every club at least once a season in the normal course of events. A decision based solely on a three-game run, without additional context, is a decision based on what this model would classify as noise.

Third, **the p5 threshold for a top-tier club at 20 games is 28 points** — approximately one win and one draw per two matches across half a season. A stretch that looks alarming in the press is still, for a top club, within the range of outcomes that the model expects to occur by chance alone roughly once every twenty seasons. The model offers no basis for a structural conclusion at that threshold.

6.4 How Quickly Do Clubs Actually Decide?

The thresholds in §6.3 raise an uncomfortable question: do clubs wait long enough for the evidence to be meaningful before acting? Across 170 mid-season managerial departures in our dataset — La Liga and the Premier League, 2013/14 to 2024/25 — the data offer one way to examine this, with important caveats.

This analysis can only see what the results say at the moment of departure. It cannot see the full set of reasons behind any decision — contract disputes, ownership changes, dressing-room dynamics, tactical disagreements, or factors entirely outside the league results. What it can ask is a narrower question: *based solely on the statistical evidence available at the time, how often do departures occur before that evidence would support a structural conclusion?*

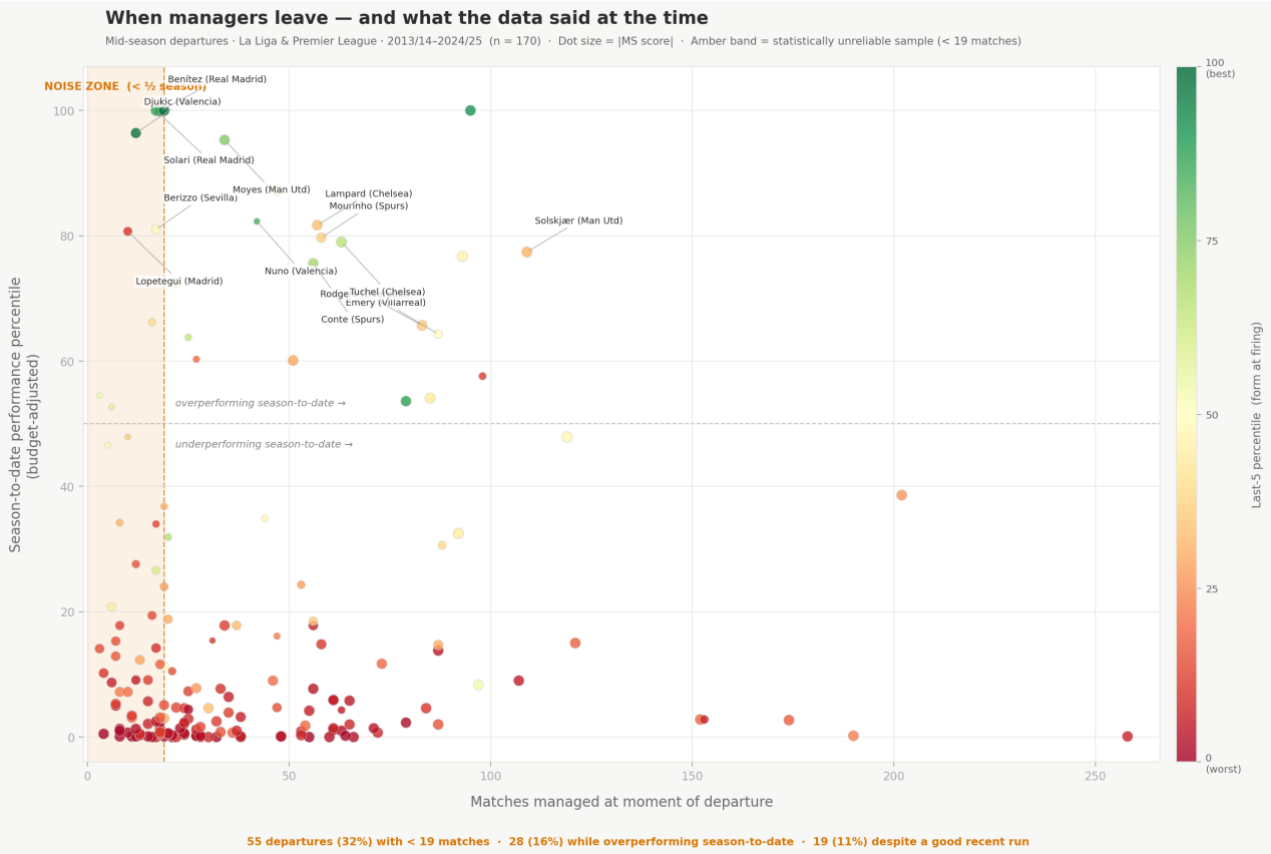

Figure 23. *Mid-season managerial departures plotted by matches managed at departure (x-axis) and season-to-date performance percentile (y-axis, budget-adjusted). Dot colour shows last-5-match percentile: green = good recent form, red = poor. Amber band marks departures with fewer than 19 matches — a sample size at which the statistical signal is particularly weak. La Liga and Premier League, 2013/14–2024/25, n = 170 permanent managers.*

Three numbers define what the results alone show. **32% of mid-season departures** occurred with fewer than 19 matches managed — less than half a season — a sample at which the model's confidence interval for any performance estimate is very wide. **16% occurred while the manager was overperforming on the season-to-date measure** — meaning the cumulative result-based evidence at the moment of departure pointed in the opposite direction. **11% happened despite a good recent run** in the last five matches. This does not mean those decisions were wrong — there may have been strong non-statistical reasons in each case. It means only that, on the evidence the model has access to, the result-based signal did not support those departures.

The amber band on the left of Figure 23 is worth examining. Cases such as Benítez, Solari, Berizzo and Djukic at Valencia appear there: departures with positive season-level MS scores after fewer than 18 league matches. At that sample size, a 95% confidence interval for any performance estimate spans an extremely wide range. Whether the non-statistical context justified those decisions is a question this analysis cannot answer.

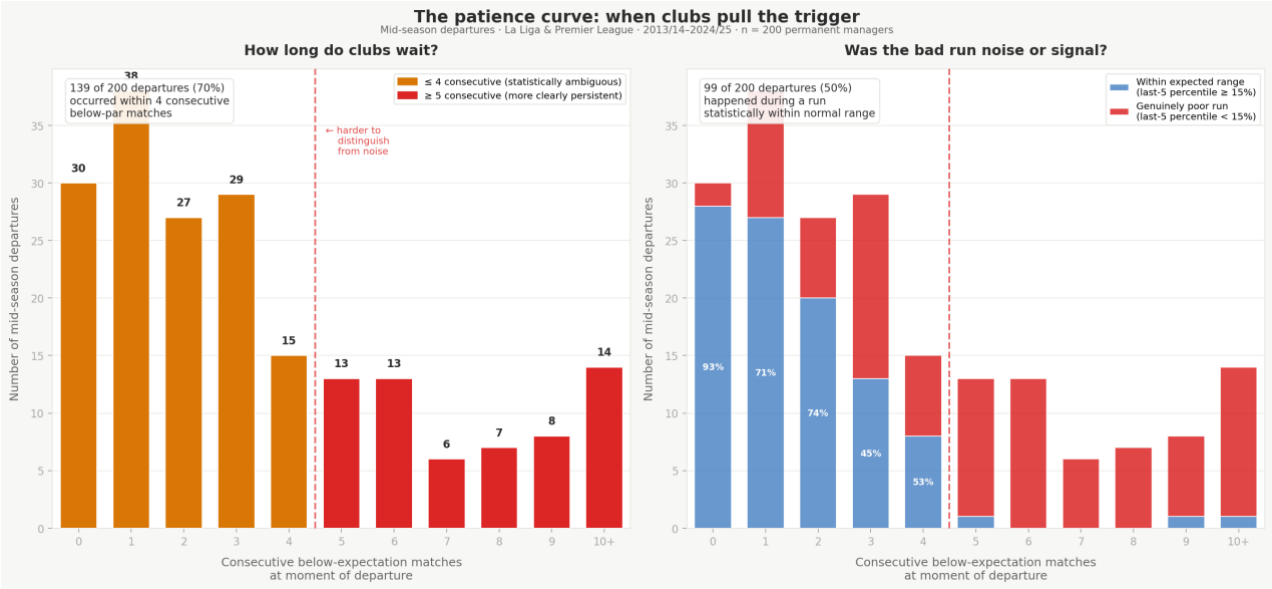

Figure 24. *Left: number of mid-season departures by consecutive below-expectation matches at the moment of departure. Amber bars = runs of 4 or fewer (statistically ambiguous at any budget tier per Figure 22); red bars = 5 or more. Right: the same departures split by whether the last-5 run was within the statistically expected range (blue) or genuinely poor (red). Percentages inside bars show the share within expected range for each run length. La Liga and Premier League, 2013/14–2024/25.*

Figure 24 maps what the results looked like at the moment of each departure. The left panel shows that **68% of mid-season departures occurred within four consecutive below-expectation matches** — before the run had extended to a length at which Figure 22 would classify it as statistically unusual for any budget tier. The right panel shows that **50% of all mid-season departures happened during runs that were within the statistically expected range for that club's budget and fixture list.** Blue and red bars — runs within range and runs outside it — appear in roughly equal proportions at every run length.

One possible reading of this pattern — and only one of several — is that departures frequently occur before the results-based evidence reaches a point where it would support a structural conclusion. If so, the decisions were driven by factors the model cannot observe. Alternatively, those factors may have been entirely sufficient on their own. What the data cannot do is distinguish between these two possibilities.

> *68% of mid-season managerial departures in our dataset occurred within four consecutive below-expectation matches — before the run reached the length at which Figure 22 would classify it as statistically unusual. Based on results alone, the evidence at the time of those departures was ambiguous.*

6.5 A Framework for the Board Table

The analysis suggests a simple two-step test for any bad run.

Step 1 — Is the run statistically unusual? Compute the percentile of the worst N-game stretch against the simulated distribution for this team's budget and fixtures. If it is above the 15th percentile, the run is within normal variance according to the model. No structural conclusion is warranted on the basis of results alone.

Step 2 — If unusual, is the season-level signal consistent? Check the season's cumulative overperformance against expectation. A statistically unusual bad run in a season where the overall MS score remains positive suggests a bad patch in an otherwise solid campaign. A statistically unusual bad run in a season where the overall MS score is also negative and trending down is more consistent with structural deterioration.

Figure 25. *A framework for the board table: interpreting a bad run of results. Step 1 tests whether the run is statistically unusual relative to the team's simulation distribution (threshold: p15). Step 2 tests whether the season-level MS signal is consistent with that reading. The four quadrants carry different action implications; real examples from the dataset are shown in each cell.*

The framework does not prescribe firing or not firing a manager. It prescribes *what the evidence actually says*, which is a prior step that appears to be routinely skipped. The model cannot see injuries, dressing-room dynamics or transfer failures. What it can see is whether the relationship between the club's resources and its results has shifted in a way that is large enough, and sustained enough, to be more than chance.

> *The practical implication is direct: a club in the top-left quadrant that fires its manager is destroying a managerial project on the basis of variance. The MS score for the season is the corrective — it integrates the full 38-match signal, not just the five worst matches.*

CALLOUT — A Real-World Test: Xabi Alonso at Real Madrid, 2025/26

⚠ Scope: La Liga only (matchweeks 1–16, 2025/26). Champions League and cup data are not included in this model. The analysis covers what the results say — not the full institutional picture.

On 12 January 2026, Real Madrid dismissed Xabi Alonso seven months into a three-year contract, one day after a 3–2 defeat to Barcelona in the Spanish Super Cup final. The club announced the departure as mutual consent; sources close to Alonso described it as a sacking.

Our model offers a precise reading of the La Liga record at the moment of dismissal. Across 16 league matches, Real Madrid had accumulated 38 points against an expected total of 29.7 — a residual of **+8.3 points**. Their worst rolling 5-game run produced 8 points, a figure that sits at the **89th percentile** of 50,000 simulated seasons for a club of their wage bill: nine in ten simulated Real Madrids had a worse bad patch than Alonso's did. The MS signal at the point of firing was firmly positive. The club were second in La Liga, four points behind Barcelona, with a model title probability of approximately one in three.

By every metric this analysis employs, Alonso was fired while statistically overperforming.

The triggers the press reported — a 5–2 derby defeat to Atlético, home losses to Celta Vigo and Getafe, a dressing-room confrontation with Vinicius Júnior, a dispute over the fitness coaching staff

— are real events. The model does not read them as evidence of structural underperformance. A team at Real Madrid's payroll level is expected to lose matches against Atlético and to drop points against mid-table opposition from time to time; that variance is already priced into the expected points total. Removing it from the residual is precisely what the expected-value framework is designed to do.

What the model cannot see is institutional friction: a breakdown of trust between manager and president, dressing-room dynamics, or a stylistic mismatch between a high-press philosophy and a squad built for a different system. Those are legitimate grounds for a decision — but they are a different argument from the one made publicly, which leaned heavily on results. The results, on the evidence available, did not support it.

Alonso's case sits in the top-right quadrant of Figure 25: no statistically unusual bad run, positive season-level MS signal, and a managerial project terminated mid-execution. Whether the non-statistical reasons were sufficient is a judgement the data cannot make. What the data can say is that the board that made this decision was not responding to a signal from the results — and that, in terms of the framework in §6.5, is precisely the scenario the analysis is designed to flag.

6.6 What the Data Cannot Tell You

The model's blind spots matter here more than anywhere else. A bad run driven by a catastrophic injury to the squad's key player, by a sudden tactical shift that the data have not yet absorbed, or by extraordinary fixture congestion will look identical in this analysis to a bad run driven by genuine managerial failure. The percentile score flags the unusual — it does not diagnose the cause.

The honest version of this framework is probabilistic, not deterministic. It says: *the probability that this run happened by chance given this team's resources is X%.* Below some threshold — 10%, 5%, whatever the board decides is their tolerance — the run deserves investigation. Above that threshold, the investigation should probably be about something other than the manager.

What twelve seasons of data do establish clearly is the base rate: **the typical club in either league has at least one 5-game run per season that would look alarming in a press conference and is entirely normal in the model.** The gap between those two descriptions is exactly the space Nassim Nicholas Taleb was writing about in *Fooled by Randomness* — and exactly the space where managerial careers are destroyed by coincidence.

> *The most consequential decision in football management may not be who to hire. It may be resisting the pressure to fire someone whose results are indistinguishable from variance — and having the analytical tools to tell the difference.*

CONCLUSIONS

What the Data Tell Us

Throughout this work we have built a predictive model calibrated on twelve seasons of La Liga and the Premier League and used it as a benchmark to measure who performs above or below their resources. We have then developed a framework — the Managerial Success Score — to classify, track and explain that overperformance over time, extended it into a within-season diagnostic for bad runs, and shown how all of it connects to the structural budget dynamics of clubs. These are seven conclusions.

1. Money Explains a Great Deal, But Not Everything

The ordered logistic regression model on the wage ratio captures the general hierarchy of both leagues well: teams with greater investment win more matches, accumulate more points and have higher title probabilities. The estimated β parameter — 0.47 in La Liga and 0.68 in the Premier League — quantifies that relationship in concrete terms: when two clubs have equal wage bills, the home side wins approximately **44–45% of matches**. When the home team doubles its wage advantage, it gains **12 percentage points in La Liga and 17 in the Premier League**. Money buys probabilities, not certainties. At match level, the wage bill alone explains only a modest fraction of outcome variance. But scale matters: at full-season level, the same model explains **68% of the variance between teams in La Liga and 52% in the Premier League** — with mean absolute errors of ±10 and ±13 points respectively. The wage bill is weakly predictive match by match, but highly predictive across a full season.

A simple model with so few parameters that remains useful for predicting and ranking results over twelve seasons is evidence that the wage bill is *the most powerful single variable available* for explaining sporting performance. But it is also evidence that management, tactics and context matter — and can be measured precisely as the difference between what actually happened and what was predicted.

2. The Two Leagues Are Structurally Different

La Liga and the Premier League share the same relationship between wage bill and outcome, but have very different dynamics. La Liga is more deterministic: the champion has always come from the top two or three wage bills, the model correctly identifies the favourite in 6 of 12 seasons, and the dispersion of overperformance between teams is relatively contained. The Premier League is more chaotic: **the model has identified the favourite correctly just twice in twelve seasons**, champions have come from wage rank #1 down to #16, and the dispersion of overperformance is almost double.

A plausible hypothesis is that greater financial balance produces more noise rather than more predictability: when wage bills are more similar, other factors — tactics, form, injuries, management — carry more relative weight. The Premier League may be a league where management makes more of a difference precisely because money makes less of one. The β parameter is 43% higher in the Premier League, meaning that when a wage gap does exist it weighs more per match — but those gaps are rarer.

3. Overperformance Is Structural, Not Noise — and Can Be Measured

The cumulative overperformance rankings across twelve seasons are not random. Assuming annual residuals are independent with standard deviation ±10 pts (La Liga) and ±13 pts (Premier League), the probability that the extreme cases are due to chance is below 10%. Villarreal, Atlético de Madrid, Liverpool, Man City, Brentford, Valencia and Man United all have structural signatures — not lucky or unlucky runs.

The Managerial Success Score (MS) makes this rigorous. It combines direction, magnitude and

statistical confidence into a single number: MS = sign(residual) × (100 − p), where p is the two-tailed probability of the cumulative residual under the null of no systematic management effect. Scores above +70 or below −70 reflect signals large enough to rule out chance at conventional significance levels; scores near zero mean the club is getting exactly what their wage bill predicts. The MS heatmaps across twelve seasons reveal not just who overperforms but *when* — Villarreal's peak during the Emery era, Man United's monotonic decline across three successive regimes, Atlético's remarkable stability above +70 in every trailing window.

4. Sustained Overperformance Drives Budget Growth

Using a lagged regression grid (window W, lag k), we find that five years of consistent overperformance explains 47% of subsequent payroll growth in La Liga and 23% in the Premier League, at a two-season lag (W=5, k=2). A single season has near-zero predictive power; the signal builds monotonically as the window lengthens from one to five years. The relationship then plateaus — five years is the optimal signal horizon.

An important secondary finding: it does not matter how overperformance is measured. MS score, normalised overperformance (%) and raw overperformance (points) all yield statistically indistinguishable R^2 values at the same sample and window. Boards respond to the scoreboard, not the underlying process. A run driven by genuine tactical superiority and one driven by a fortunate draw are rewarded at the same rate. This makes the next conclusion — about *what actually produces* sustained overperformance — more urgent, not less.

5. The Differentiator Is Managerial Continuity, Not Budget Level

Across 127 five-year windows in both leagues, the clubs that combine sustained overperformance with strong subsequent payroll growth — the top-right quadrant — span the full budget range: from Getafe at €27M to Man City at £156M. Sociedad and Betis sit alongside Liverpool and Tottenham. Budget size is not the differentiator. The one attribute shared without exception across all nine top-right clubs is a coherent, multi-year coaching project.

The bottom-left clubs — Valencia, Celta, Crystal Palace — are mid-budget clubs, not financially distressed. What they share is managerial instability: seven managers in eight seasons at Valencia, six at Celta, five in six seasons at Crystal Palace. No project was allowed to take root. A Fisher exact test on the 2×2 contingency of managerial stability versus quadrant membership yields p=0.019 for the top-right and p=0.025 for the bottom-left. The three outliers — Barcelona (financial crisis forced wage cuts), Man United (Glazers kept spending despite failure), Newcastle (PIF takeover) — sit conspicuously off the regression line precisely where an external shock decoupling budget from performance would predict. They are the strongest confirmation of the rule.

A methodological note that the data cannot resolve: the direction of this relationship may run in both directions. Well-governed clubs likely produce both stable managerial tenures *and* revenue growth independently — institutional quality enabling both outcomes simultaneously rather than continuity mechanically causing commercial growth. The association is robust; the causal arrow requires careful interpretation. What the analysis demonstrates is that the two conditions — sustained overperformance and managerial continuity — are closely linked in the data across both leagues. Whether that is because continuity produces results, results protect tenure, or a common institutional factor drives both is a question the current methodology cannot fully settle.

6. The Strategic Implication: Commit to the Right Coach

The data establish a feedback loop: sustained overperformance → commercial and revenue growth → larger wage bill → sustained overperformance. Every club in this analysis that successfully entered and maintained that loop did so by building a coherent multi-year coaching project. Every club that failed to enter it — or was knocked out of it — did so through managerial instability, regardless of budget level. The mechanism is not subtle: it takes at least three seasons for the MS signal to strengthen materially, and five seasons for the budget growth feedback to operate at full strength. Clubs that change manager every 18 months are structurally incapable

of entering the loop.

This has a direct implication for club strategy. Investing in the largest possible wage bill is necessary but insufficient. The clubs that have built durable competitive advantage — in both leagues, at all budget levels — have done three things consistently: they identified a coaching project aligned with their structural resources; they gave that project the institutional authority to operate (control over recruitment, squad culture, tactical identity); and they protected it from short-term results noise long enough for the signal to become structural. The practical question for any club board is not *how much to spend on wages*, but *who to give the keys to* — and whether the institution is genuinely willing to support them.

7. Most Bad Runs Are Noise — and the Model Can Tell the Difference

Chapter 6 extends the analytical framework from seasons to matches. For every team-season in the dataset we simulated 50,000 full seasons, preserving the actual fixture sequence and wage ratios, and extracted the distribution of worst rolling N-game stretches. The central finding is counterintuitive in its magnitude: a top-quartile club going winless across five games is expected to happen by pure chance in roughly **30% of simulated seasons**. Three consecutive defeats is the median worst run for every budget tier. The threshold for a run to be genuinely statistically unusual — below the 5th percentile of simulations — is considerably higher than typical board-meeting narratives imply.

Combining the rolling-run percentile with the season-level MS score produces a two-dimensional diagnostic. The bottom-left quadrant — extreme bad runs combined with a negative MS score — is the firing zone that is justified; Valencia 2015/16, Chelsea 2022/23 and Tottenham 2023/24 all sit there. The top-left quadrant — extreme bad run but positive season-level MS — is the firing zone that is a mistake. The top-right — normal runs, positive MS — is the noise zone. Real Madrid 2025/26 under Xabi Alonso sat in the top-right at the point of dismissal: 38 points from 16 La Liga matches against an expected 29.7, with their worst 5-game run at the 89th percentile of simulations. The decision may have had legitimate non-statistical grounds; the results did not provide them.

The practical implication is a two-step board test. First: is the run statistically unusual for this club's budget and fixture list? If not, the conversation should not be about the manager. Second: if it is unusual, is the season-level MS signal consistent? A negative answer to both means the club is acting on variance. The cases examined in Chapter 6 suggest this pattern is common — the framework implies that a significant share of managerial changes in both leagues are made in response to runs that the model would classify as normal variance. That implication warrants further systematic study; the data here establish the diagnostic, not a census of decisions.

APPENDIX

Extended Analysis: Atlético de Madrid, the Big Three and La Liga 2025/26

The following sections extend the analysis for readers interested in specific applications: the statistical interpretation of Atlético de Madrid's title record (A.1), head-to-head performance among La Liga's Big Three (A.2), and the model's real-time projection for La Liga 2025/26 (A.3). These sections use the same model and metrics as the main text.

APPENDIX A.1

What Can We Reasonably Expect from Atlético de Madrid?

In La Liga, beyond the competition between Real Madrid and Barcelona, one of the most recurring debates in the sporting press is how much can reasonably be demanded of Atlético de Madrid in terms of titles. Are they genuinely a contender on equal terms with the two big clubs? Can they be reproached for not winning more? The analysis in the previous section already showed that Atlético consistently outperform their wage bill. The question here is more demanding: what does that mean in terms of league titles?

Figure 14 shows the title probabilities the model assigns to each of the Big Three in each season since 2013/14, along with expected and actual points. The asymmetry is structural and persistent: Barcelona and Real Madrid compete against each other with probabilities that range between 30% and 70% depending on the year, while Atlético rarely exceed 17%. It is not that Atlético play poorly; it is that their resources do not generate comparable title probabilities.

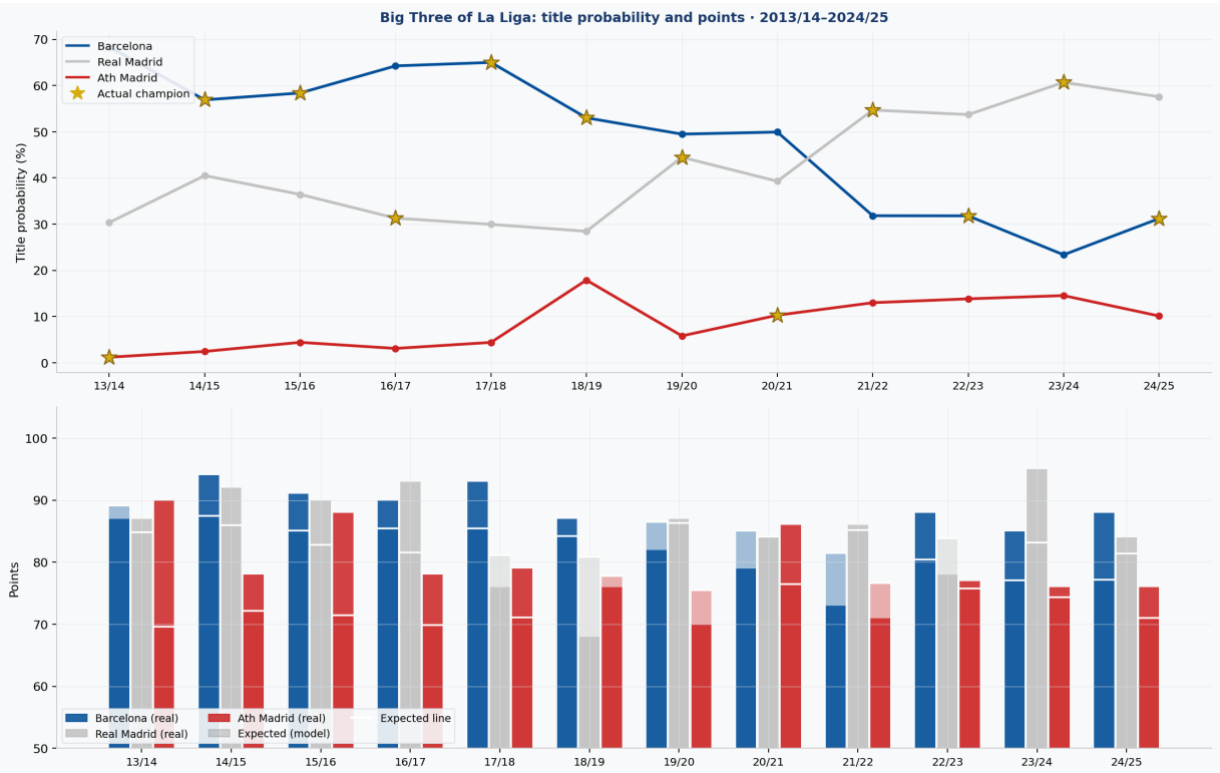

Figure 14. *Upper panel: model-assigned title probability at the start of each season for La Liga's Big Three (2013/14–2024/25). The star (★) marks the actual champion. Lower panel: expected points (transparent bar) and actual points (solid bar) for the same teams and period. The inner white line marks the model's expectation.*

This context makes what happened in **2013/14** remarkable. The model assigned Atlético a title probability of **1.2%** — Barcelona were overwhelming favourites at 68%, Madrid at 31%. Atlético did not merely win: they did so accumulating **90 points**, 21 above the 69 the model expected given their wage bill. It was the largest positive deviation by any team in any season of the period, in either of the two leagues analysed. The 2013/14 title was not a mild statistical surprise; it was the most improbable result of the entire period.

The **2020/21 title** is different in nature. Atlético's annual probability had grown to 10.3% — their fifth most favourable year of the period — and the cumulative probability of having won at least one league over those 8 seasons had already reached 40.8%. Winning at that point was not an anomaly; it was a reasonably expected materialisation. Furthermore, the season involved a confluence of factors: Barcelona finished 5.2 points below their expectation, Madrid barely exceeded theirs by 0.7 points, while Atlético exceeded theirs by 10.3. Atlético's two titles are, statistically, completely different in nature.

> *The 2013/14 title occurred with a cumulative probability of 1.2% — the most improbable result of the period across both leagues. The 2020/21 title came when that cumulative probability had already reached 40.8%. These are not two versions of the same phenomenon: they are statistically distinct events that the narrative of 'one title every ten years' artificially collapses into a single story.*

Figure 15 illustrates both dimensions. The left panel shows the cumulative probabilities of winning 0, 1, 2 or more leagues over the period. After 12 seasons, the most likely outcome is that Atlético would have won exactly 1 league (39.2% of simulations), followed by 0 leagues (34.0%). Winning 2 occurs in 19.8% of scenarios — Atlético are in that group, at the upper but not extraordinary end of the distribution. Winning 3 leagues in this period would have been a result with only a 7.0% probability; publicly demanding it is equivalent to expecting what fails to happen 94% of the time.

The right panel puts into perspective the objective the club itself has declared as the minimum acceptable: finishing in the top three to guarantee Champions League qualification. For each season, the probability distribution by finishing position is shown, and the circle on each curve indicates where the team actually finished. Atlético's average probability of finishing in the top three is **71%** per year — well short of the 93–99% that Real Madrid and Barcelona have in any given year. In 2016/17, that probability fell to 49%: practically a coin flip. For Atlético, finishing third is not a comfortable floor; it is a target with a structural failure margin of ~30% per year according to the model. In practice, the club has missed that target only once during the period (4th in 2023/24), which reflects management consistently above expectation.

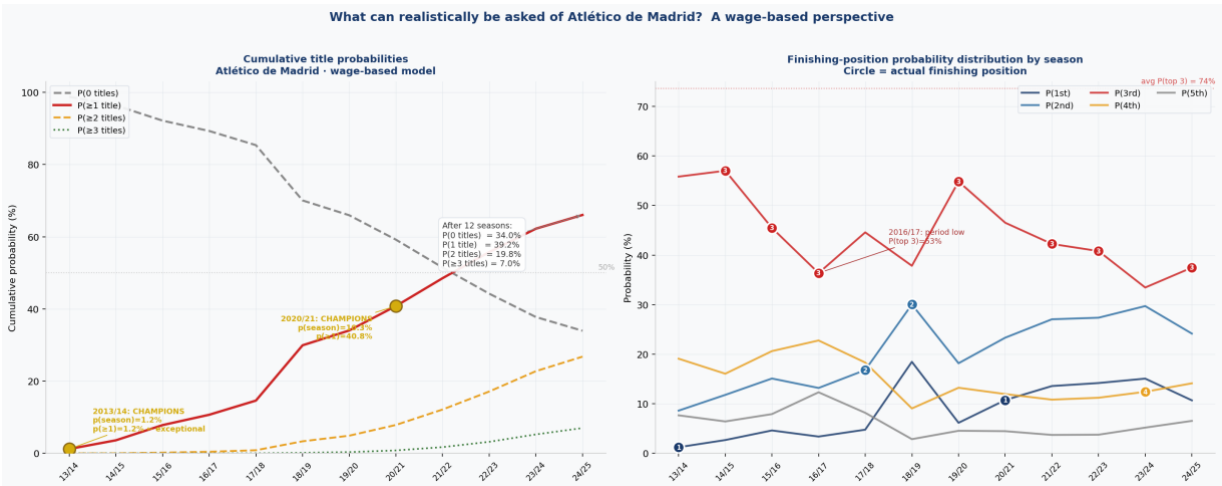

Figure 15. *Left panel: cumulative probabilities of winning 0, 1, 2 or more league titles for Atlético de Madrid over the period 2013/14–2024/25, estimated via Monte Carlo simulation (n = 300,000). Golden markers indicate the two seasons in which Atlético won the title (2013/14 and 2020/21). Right panel: finishing-position probability distribution (1st to 5th) by season. The coloured circle on each curve indicates the actual position achieved.*

The conclusion drawn from the analysis is twofold. First, winning La Liga requires Atlético to simultaneously exceed their own expectations and for the two big clubs to fall short of theirs — a conjunction the model suggests should occur at most once every twelve seasons. Second, the objective of consistently qualifying for the Champions League is in itself a management achievement, not a result guaranteed by the wage bill. The media debate about whether Atlético win enough frequently starts from comparing the club with Real Madrid and Barcelona: a comparison that wage data makes structurally invalid.

APPENDIX A.2

La Liga's Big Three: Head-to-Head and Beyond

Figure 16 organises the matches of Real Madrid, Barcelona and Atlético de Madrid in a 3×3 matrix: each row is a team and each column a rival context — the two opposing big sides, and the rest of the league. Within each cell, three groups are shown (home, away, total). The solid bar shows actual results; the hatched bar shows the model prediction. Percentages indicate the actual win rate and, when the deviation between actual and predicted exceeds 8 percentage points, the difference is annotated in the margin.

Figure 16. *Matrix of actual results (solid bar) vs. model predictions (hatched bar) for La Liga's Big Three, broken down by opponent (columns) and venue context (Home / Away / Total). Percentages show the actual win rate. Margin annotations flag deviations above 8 percentage points between actual and predicted win rates. Full seasons 2013/14–2024/25, n = 24 matches per head-to-head fixture and n = 408 for the 'vs Rest' context.*

An important detail about sample size: the head-to-head cells (n = 12 per home/away context, n = 24 in total) have **high individual variance** due to the small number of matches — a single different result is enough to shift the rate by 10 points. The most robust readings are from the 'vs Rest' column (n = 340 per team), and the head-to-head patterns should be interpreted as tendencies, not as statistically definitive findings.

Real Madrid: Dominant Against the Rest, Troubled by Atlético

Real Madrid's right-hand column is the most predictable in the entire matrix: **79% wins at home and**

64% away against the rest of the league, virtually identical to what the model expected (77% and 60%). In non-elite fixtures, Madrid performs with almost exactly the efficiency their wage bill predicts.

The surprise is in the middle column: against Atlético, Madrid **win only 30% of matches** when the model predicted 44% — a deviation of −14 percentage points in win rate. What is most striking is not the low win rate itself, but the distribution: **55% of matches end in a draw**, well above the 28% expected. Atlético do not beat Madrid, but they systematically steal a point rather than losing. This is the Simeone effect visible in the data: well-organised defensive teams convert expected defeats into draws.

Against Barcelona the balance is more even than wage bills suggest: 40% wins for Madrid overall, slightly above the 36% expected. El Clásico remains an open fixture that the model cannot predict with confidence — the wage bills are similar enough for other factors to dominate.

Barcelona: The Side That Does Not Lose Its Home Advantage Away

Barcelona's most striking figure is in the 'vs Atlético' column: **60% wins both at home and away**, a deviation of +16 pp above the overall prediction. Barça have been the most dominant side in this rivalry over the period analysed, winning at a rate far above what any wage-based analysis would anticipate.

Also notable is the limited penalty for playing away: Barcelona achieve **62% wins as the away side** against the rest of the league — compared with 80% at home — a drop of only 18 points that the model underestimated (predicted 59%). Barça travel well. Against Madrid, however, the story reverses curiously: **they lose 40% of home fixtures** (only 30% wins), yet win 60% away. An asymmetry that is difficult to attribute to structural factors and more likely related to the specific dynamics of matches in particular seasons.

Atlético de Madrid: A Fortress at Home, a Wall Against the Big Two

Atlético's row reveals two completely different faces. Against the rest of the league, Atlético are as efficient at home as Madrid and Barcelona — **79% home wins** — but with a wage bill that, according to the model, only justifies 69%. That is +10 pp of overperformance at home, the best record among the Big Three in that context, and the numerical expression of what Section 5 already showed at the cumulative level.

Against Madrid and Barcelona, the picture changes radically: **only 15% wins** in each rivalry. Here, however, Atlético show their greatest distinctiveness: in matches against Madrid they accumulate **55% draws**, and against Barcelona 25%. The message is clear — Atlético cannot compete head-to-head with the highest wage bills in terms of wins — but they have developed a style that converts those matches into something closer to a guaranteed point than a certain defeat. In the context of their resources, that is management.

> *The Big Three's pattern suggests that elite head-to-head fixtures follow a different logic from the rest: the wage gap between the three is small in relative terms, and other factors — tactics, current form, specific motivation — carry more weight. The model captures this implicitly: its predictions for these fixtures are almost always close to 33-33-33, which is itself an acknowledgement of maximum uncertainty.*

APPENDIX A.3

La Liga 2025/26: What the Model Says with 12 Matchweeks to Go

How to Read the Figures

Figure 17 shows the current La Liga standings after 26 matchweeks, combined with the model's projection for the end of the season. The solid bar shows actual points earned; the shaded extension shows the 90% confidence interval for the final projection (mean ± 1.64σ); and the diamond shows the projected mean. Each team's wage rank appears on the left. Bar colour indicates whether the team is performing above (green) or below (red) its wage expectation, with the deviation annotated when it exceeds ±5 points.

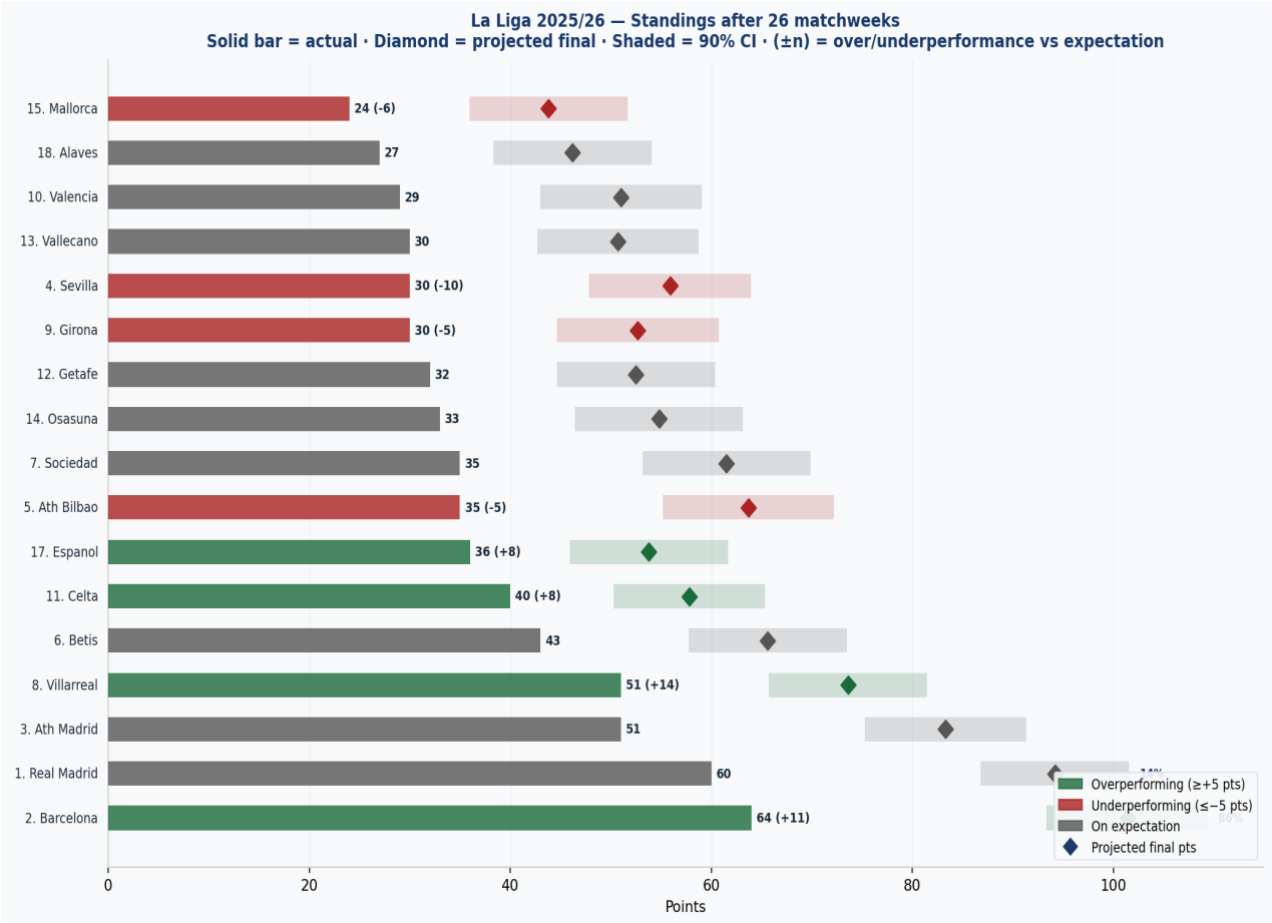

Figure 17. *La Liga 2025/26: actual standings after 26 matchweeks (solid bar), projected final points (diamond + 90% confidence interval). Colour indicates overperformance relative to the model's wage-based expectation. Each team's wage rank is shown on the left.*

The Title Race: Barcelona Favourites, Madrid Without Margin for Error

With 26 matchweeks played, the model assigns a title probability of 66.4% to Barcelona and 33.5% to Real Madrid. The 4-point gap in the table translates into a 33-percentage-point advantage in probability — a gap that reflects not only the current scoreline but the reduced variance remaining: with 12 matchweeks to play, there is less room to reverse the situation.

Barcelona reach this position from the **second-largest wage bill in the league**, with an overperformance of +11 points above expectation. In other words: the model would have predicted 53 points for Barça at this stage given their wage profile; they have 64. **Madrid, with the largest**

wage bill, overperform by only +4.5 points — performing well, but within what is expected for their investment. The story of the season so far is that the team with the second-largest wage bill is performing like the first.

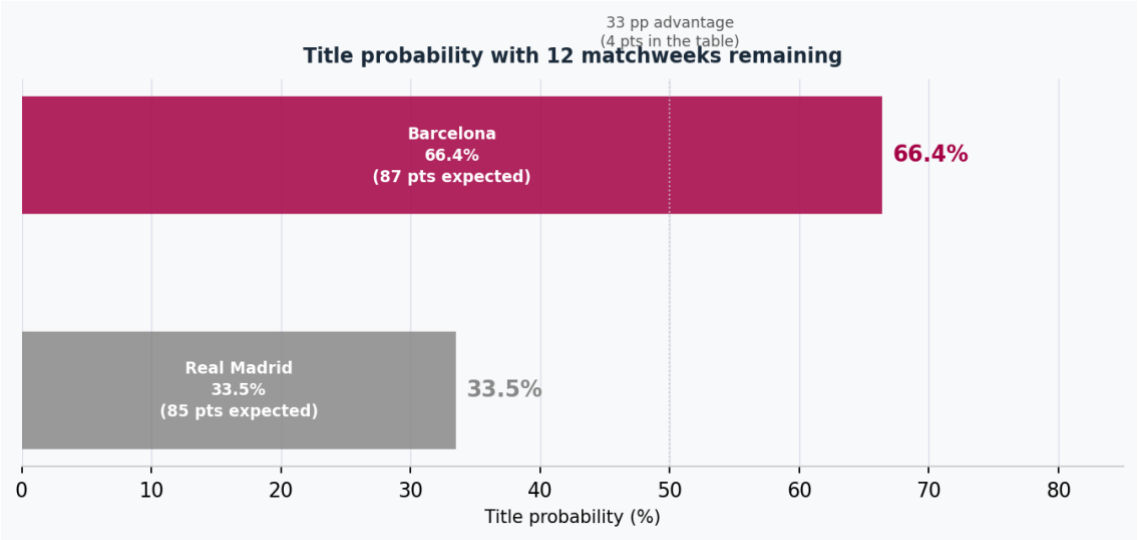

Figure 18. *Title probability in La Liga 2025/26 with 12 matchweeks remaining, from the Monte Carlo simulation updated with results from the first 26 matchweeks. The dashed line marks the 50% threshold.*

The Season's Surprises: Villarreal and Espanyol

Beyond the title race, two teams stand out for their overperformance at the top of the table. Villarreal have accumulated **+13.8 points above expectation** — the league's highest overperformance — and sit fourth despite having the seventh-largest wage bill. This is not an isolated statistical surprise: as seen in Section 5, Villarreal are the team with the greatest cumulative overperformance in La Liga over the 2013/14–2024/25 period. What they are doing this season is consistent with their track record.

The case of **Espanyol** is even more striking: with the **seventeenth-largest wage bill in the league** — close to the relegation zone in financial terms — they have accumulated 36 points and an overperformance of +8. They are performing like a side challenging for Europe with promoted-club resources. At the opposite extreme, Sevilla continues the negative trend already documented in the historical record: with the eighth-largest wage bill, they have accumulated −8 points below expectation and sit thirteenth.

The Relegation Zone: The Model Has Its Candidates

At the bottom, the picture is clear: Oviedo — newly promoted — have the league's worst negative overperformance (−10 points), with 17 actual points against the 27 expected. They are performing as the weakest team in the league in relative terms. **Mallorca** (−5) and **Levante** (−5) also appear in the relegation zone below their wage-based expectation, while **Alavés** are broadly in line with what the model predicted.

> *An important methodological note: the final projection assumes that the wage bills for the rest of the season are equivalent to those at the start. If any team has changed manager or suffered serious injuries, the model does not capture this. The projections are a probabilistic reference, not a deterministic prediction.*

ANNEX

Wage Bill and Points Data

The following tables contain the underlying data for the analysis. La Liga wage bills come from Capology and are expressed in millions of euros (M€); Premier League wage bills are expressed in millions of pounds sterling (M£). All figures are rounded to the nearest integer. Predicted points (P) correspond to the Monte Carlo simulation mean; actual points (A) are the official points at the end of each season. A dash (—) indicates the team did not compete in that division that season.

Table A1. *La Liga wage bills 2013/14–2024/25 (M€, rounded to nearest integer).*

La Liga — Annual wage bill (M€)												
Team	**13/**	**14/**	**15/**	**16/**	**17/**	**18/**	**19/**	**20/**	**21/**	**22/**	**23/**	**24/**
Barcelona	186	171	180	262	284	314	357	327	264	240	201	220
Real Madrid	135	152	152	199	211	249	352	306	341	305	302	288
Ath Madrid	50	60	73	94	111	203	167	181	191	176	170	150
Sevilla	25	26	34	70	69	76	71	72	81	82	85	73
Valencia	32	31	52	71	61	75	75	74	65	48	37	40
Ath Bilbao	19	24	26	37	37	53	57	70	63	61	70	73
Villarreal	15	26	23	31	32	48	51	66	70	55	60	55
Sociedad	15	11	21	23	32	37	40	47	49	54	69	59
Betis	15	—	21	18	27	35	35	39	47	48	54	66
Celta	9	14	22	41	45	69	36	39	37	30	33	37
Espanol	—	—	15	33	31	30	37	—	30	34	—	27
Girona	—	—	—	—	14	18	—	—	—	23	34	48
Getafe	15	15	13	—	14	19	26	26	36	42	35	36
Leganes	—	—	—	17	22	36	29	—	—	—	—	19
Mallorca	—	—	—	—	—	—	12	—	21	23	26	31
La Coruna	—	—	14	30	21	—	—	—	—	—	—	—
Cadiz	—	—	—	—	—	—	—	17	19	20	29	—
Levante	11	13	17	—	32	37	20	18	19	—	—	—
Alaves	—	—	—	12	16	22	22	24	22	—	17	26
Granada	10	12	17	24	—	—	18	26	25	—	24	—
Malaga	17	17	16	23	24	—	—	—	—	—	—	—
Osasuna	11	—	—	14	—	—	14	20	17	20	23	32
Vallecano	10	12	13	—	—	19	—	—	11	18	26	34
Valladolid	7	—	—	—	—	16	15	23	—	22	—	25
Las Palmas	—	—	8	14	20	—	—	—	—	—	18	28
Huesca	—	—	—	—	—	15	—	17	—	—	—	—
Eibar	—	7	13	18	18	19	19	18	—	—	—	—
Espanyol	14	18	—	—	—	—	—	—	—	—	—	—
Elche	11	5	—	—	—	—	—	13	20	20	—	—
Almeria	8	5	—	—	—	—	—	—	—	15	19	—
Deportivo	—	11	—	—	—	—	—	—	—	—	—	—
Sp Gijon	—	—	6	13	—	—	—	—	—	—	—	—
Cordoba	—	8	—	—	—	—	—	—	—	—	—	—

Table A2. *Premier League wage bills 2013/14–2024/25 (M£, rounded to nearest integer).*

Premier League — Annual wage bill (M£)												
Team	**13/**	**14/**	**15/**	**16/**	**17/**	**18/**	**19/**	**20/**	**21/**	**22/**	**23/**	**24/**
Man United	97	143	123	152	163	150	164	192	234	242	196	171
Man City	114	123	127	135	136	151	156	144	169	199	197	226
Chelsea	106	105	100	99	122	172	129	163	187	227	157	171
Arsenal	74	95	97	102	161	138	119	153	118	133	171	170
Liverpool	66	82	94	91	97	122	116	132	155	171	142	148
Tottenham	55	68	53	71	80	91	106	136	110	117	134	121
Aston Villa	34	33	37	—	—	—	51	65	85	109	114	140

Premier League — Annual wage bill (M£)												
Team	13/	14/	15/	16/	17/	18/	19/	20/	21/	22/	23/	24/
Everton	53	50	55	68	84	85	93	103	108	86	81	74
West Ham	47	45	53	55	68	84	77	69	78	89	105	103
Crystal Palace	19	32	54	72	68	74	74	90	72	70	74	77
Nott'm Forest	—	—	—	—	—	—	—	—	—	77	74	66
Leicester	—	33	38	63	66	78	68	80	82	85	—	67
Newcastle	36	32	47	—	38	47	65	60	74	92	88	110
Fulham	33	—	—	—	—	50	—	56	—	62	68	72
Wolves	—	—	—	—	—	41	50	56	63	68	62	82
Southampton	28	44	44	57	67	76	56	63	59	62	—	54
Brighton	—	—	—	—	39	43	48	53	50	45	67	68
Stoke	34	37	42	54	51	—	—	—	—	—	—	—
Sunderland	42	39	46	48	—	—	—	—	—	—	—	—
Watford	—	—	31	47	49	49	57	—	42	—	—	—
Bournemouth	—	—	27	34	42	54	45	—	—	51	57	54
West Brom	35	35	42	41	58	—	—	37	—	—	—	—
Leeds	—	—	—	—	—	—	—	32	37	61	—	—
Swansea	21	30	41	40	48	—	—	—	—	—	—	—
Hull	21	37	—	42	—	—	—	—	—	—	—	—
QPR	—	45	—	—	—	—	—	—	—	—	—	—
Ipswich	—	—	—	—	—	—	—	—	—	—	—	41
Brentford	—	—	—	—	—	—	—	—	29	38	47	45
Middlesbrough	—	—	—	39	—	—	—	—	—	—	—	—
Burnley	—	16	—	25	32	44	44	36	41	—	41	—
Cardiff	28	—	—	—	—	33	—	—	—	—	—	—
Norwich	28	—	33	—	—	—	27	—	32	—	—	—
Huddersfield	—	—	—	—	26	35	—	—	—	—	—	—
Sheffield United	—	—	—	—	—	—	24	21	—	—	35	—
Luton	—	—	—	—	—	—	—	—	—	—	26	—

Table A3. *Predicted (P) and actual (A) points · La Liga 2013/14–2024/25.*

La Liga — Predicted vs actual points																								
	13/		14/		15/		16/		17/		18/		19/		20/		21/		22/		23/		24/	
Team	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R
Barcelona	88	87	87	94	84	91	85	90	84	93	83	87	86	82	84	79	81	73	80	88	76	85	76	88
Real Madrid	84	87	85	92	82	90	81	93	80	76	80	68	86	87	83	84	84	86	83	78	82	95	81	84
Ath Madrid	69	90	72	78	71	88	69	78	71	79	77	76	75	70	76	86	76	71	75	77	74	76	70	76
Sevilla	58	63	58	76	58	52	65	72	63	58	61	59	61	70	61	77	62	70	63	49	62	41	58	41
Valencia	62	49	61	77	65	44	65	46	61	73	61	61	62	53	61	43	59	48	54	42	49	49	48	46
Ath Bilbao	53	70	57	55	54	62	54	63	52	43	56	53	58	51	61	46	58	55	58	51	59	68	58	70
Villarreal	49	59	58	60	52	64	51	67	50	61	54	44	56	60	59	58	60	59	56	64	57	53	53	70
Sociedad	49	59	44	46	51	48	46	64	50	49	50	50	52	56	54	62	54	62	56	71	59	60	55	46
Betis	50	25	—	—	50	45	42	39	47	60	48	50	50	41	51	61	54	65	54	60	55	57	57	60
Celta	40	49	47	51	51	60	56	45	56	49	60	41	50	37	51	53	49	46	46	43	47	41	47	55
Espanol	—	—	—	—	45	43	52	56	50	49	46	53	51	25	—	—	46	42	48	37	—	—	42	42
Girona	—	—	—	—	—	—	—	—	37	51	38	37	—	—	—	—	—	—	42	49	47	81	51	41
Getafe	49	42	49	37	43	36	—	—	37	55	39	59	45	54	44	38	49	39	51	42	48	43	47	42
Leganes	—	—	—	—	—	—	41	35	44	43	49	45	46	36	—	—	—	—	—	—	—	—	36	40
Mallorca	—	—	—	—	—	—	—	—	—	—	—	—	33	33	—	—	41	39	42	50	43	40	44	48
La Coruna	—	—	—	—	43	42	50	36	43	29	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Cadiz	—	—	—	—	—	—	—	—	—	—	—	—	—	—	38	44	39	39	40	42	45	33	—	—
Levante	44	48	47	37	47	32	—	—	50	46	50	44	41	49	39	41	39	35	—	—	—	—	—	—
Alaves	—	—	—	—	—	—	36	55	39	47	41	50	43	39	43	38	41	31	—	—	37	46	41	42
Granada	43	41	45	35	47	39	47	20	—	—	—	—	40	56	44	46	43	38	—	—	42	21	—	—
Malaga	51	45	51	50	46	48	46	46	45	20	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Osasuna	43	39	—	—	—	—	39	22	—	—	—	—	35	52	40	44	37	47	40	53	41	45	45	52
Vallecano	42	43	46	49	43	38	—	—	—	—	39	32	—	—	—	—	32	42	38	49	43	38	46	52
Valladolid	37	36	—	—	—	—	—	—	—	—	36	41	37	42	43	31	—	—	41	40	—	—	40	16
Las Palmas	—	—	—	—	35	44	39	39	42	22	—	—	—	—	—	—	—	—	—	—	37	40	42	32
Huesca	—	—	—	—	—	—	—	—	—	—	35	33	—	—	38	34	—	—	—	—	—	—	—	—
Eibar	—	—	38	35	43	43	42	54	40	51	39	47	40	42	38	30	—	—	—	—	—	—	—	—
Espanyol	48	42	52	49	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Elche	44	40	33	41	—	—	—	—	—	—	—	—	—	—	34	36	40	42	40	25	—	—	—	—
Almeria	39	40	33	32	—	—	—	—	—	—	—	—	—	—	—	—	—	—	36	41	38	21	—	—
Deportivo	—	—	43	35	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Sp Gijon	—	—	—	—	32	39	37	31	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Cordoba	—	—	40	20	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table A4. *Predicted (P) and actual (A) points · Premier League 2013/14–2024/25.*

Premier League — Predicted vs actual points																								
	13/		14/		15/		16/		17/		18/		19/		20/		21/		22/		23/		24/	
Team	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R	P	R
Man United	74	64	80	70	74	66	76	69	75	81	71	66	74	66	76	74	79	58	77	75	74	60	68	42
Man City	77	86	76	79	74	66	73	78	71	100	71	98	73	81	69	86	72	93	72	89	74	91	75	71
Chelsea	76	82	72	87	69	50	65	93	68	70	75	72	68	66	72	67	74	74	75	44	68	63	68	69
Arsenal	67	79	70	75	68	71	66	75	75	63	69	70	66	56	71	61	63	69	62	84	70	89	68	74
Liverpool	64	84	66	62	67	60	63	76	62	75	66	97	65	99	67	69	70	92	68	67	66	82	65	84
Tottenham	59	69	61	64	53	70	57	86	58	77	59	71	63	59	68	62	62	71	59	60	64	66	60	38
Aston Villa	47	38	43	38	44	17	—	—	—	—	—	—	45	35	49	55	55	45	57	61	60	68	63	66
Everton	58	72	54	47	53	47	55	61	59	49	57	54	60	49	61	59	61	39	51	36	52	48	47	48
West Ham	55	40	51	47	52	62	50	45	53	42	57	52	55	39	51	65	53	56	52	40	58	52	55	43
Crystal Palace	34	45	42	48	53	42	57	41	54	44	53	49	54	43	57	44	51	48	46	45	49	49	48	53
Nott'm Forest	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	48	38	49	36	44	65
Leicester	—	—	43	41	44	81	54	44	53	47	55	52	52	62	54	66	54	52	51	34	—	—	44	25
Newcastle	49	49	43	39	49	37	—	—	39	44	42	45	51	44	47	45	51	49	53	71	54	60	57	66
Fulham	47	32	—	—	—	—	—	—	—	—	44	26	—	—	46	28	—	—	43	52	47	47	46	54
Wolves	—	—	—	—	—	—	—	—	—	—	39	57	45	59	46	45	48	51	45	41	45	46	50	42
Southampton	42	56	50	60	48	63	51	46	53	36	54	39	47	52	48	43	46	40	43	25	—	—	40	12
Brighton	—	—	—	—	—	—	—	—	40	40	40	36	43	41	44	41	42	51	36	62	47	48	45	61

Premier League — Predicted vs actual points																								
	13/		14/		15/		16/		17/		18/		19/		20/		21/		22/		23/		24/	
Stoke	47	50	46	54	47	51	50	44	46	33	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Sunderland	52	38	47	38	49	39	47	24	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Watford	—	—	—	—	39	45	46	40	45	41	43	50	48	34	—	—	38	23	—	—	—	—	—	—
Bournemouth	—	—	—	—	36	42	38	46	42	44	46	45	42	34	—	—	—	—	38	39	43	48	39	56
West Brom	48	36	45	44	46	43	43	45	49	31	—	—	—	—	36	26	—	—	—	—	—	—	—	—
Leeds	—	—	—	—	—	—	—	—	—	—	—	—	—	—	33	59	35	38	43	31	—	—	—	—
Swansea	36	42	41	56	46	47	42	41	45	33	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Hull	36	37	46	35	—	—	44	34	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
QPR	—	—	51	30	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Ipswich	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	33	22
Brentford	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	30	46	32	59	38	39	35	56
Middlesbrough	—	—	—	—	—	—	42	28	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Burnley	—	—	27	33	—	—	31	40	35	54	41	40	42	54	36	39	37	35	—	—	35	24	—	—
Cardiff	42	30	—	—	—	—	—	—	—	—	34	34	—	—	—	—	—	—	—	—	—	—	—	—
Norwich	42	33	—	—	40	34	—	—	—	—	—	—	30	21	—	—	32	22	—	—	—	—	—	—
Huddersfield	—	—	—	—	—	—	—	—	31	37	36	16	—	—	—	—	—	—	—	—	—	—	—	—
Sheffield United	—	—	—	—	—	—	—	—	—	—	—	—	28	54	24	23	—	—	—	—	32	16	—	—
Luton	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	26	26	—	—

METHODOLOGICAL NOTE

Sources, Tools and Use of Artificial Intelligence

Data Sources

Match result data come from Football-Data.co.uk, which publishes historical records for La Liga and the Premier League in CSV format from the 1993/94 season. For this analysis, the seasons from 2013/14 to 2024/25 were used, covering a total of 4,448 matches (La Liga) and 4,560 matches (Premier League). Squad wage bill data come from Capology and from annual reports published by the clubs themselves and the relevant federations. Figures for the ongoing 2025/26 season correspond to data available as of March 2026.

Analysis Tools

The statistical model and simulations were implemented in Python, using the statsmodels library (ordered logistic regression), NumPy and SciPy (Monte Carlo simulation, statistical tests) and Matplotlib (visualisations). All charts in the document were generated programmatically from the original data. The final document was produced using the Node.js docx library.

Use of Artificial Intelligence

This analysis was produced with the assistance of the Claude artificial intelligence assistant (Anthropic), in an iterative collaborative process between the author and the model. The use of AI covered three main dimensions:

> Technical implementation. Claude wrote and debugged the Python code for model estimation, Monte Carlo simulation and chart generation, as well as the Node.js code for document composition. The author defined the analytical objectives, reviewed the results and validated the consistency of each step.
>
> Writing and interpretation. The texts for each section were drafted by Claude based on the author's instructions and criteria, who reviewed, corrected and approved each paragraph. The interpretation of results, the selection of highlighted cases and the conclusions reflect the author's analytical judgement.
>
> Design and layout. The visual structure of the document, the colour palette, chart design and cover page were defined collaboratively. Claude proposed options and implemented the adjustments indicated by the author.

At no point did the AI model make analytical decisions autonomously. All methodological choices — the selection of the ordered logit model, the use of Monte Carlo simulation, the approach to quantifying overperformance through chance probability, or the selection of teams and periods analysed — were defined and validated by the author. The AI acted as a tool for implementation and drafting, not as an independent analytical agent.

Model used: *Claude Sonnet (Anthropic), March 2026.*

Comments and contact: whatmoneycantbuyfootball@gmail.com